Employee Attrition AnalysiS

Insights Into Management, Career Progression & Retention

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Dataset Introduction

IBM HR Analytics Dataset

The dataset used in this analysis is the IBM HR Analytics Employee Attrition and Performance dataset, publicly available on Kaggle. It contains 1,470 employee records across 35 variables capturing demographic information, compensation details, job characteristics, satisfaction scores, and attrition status.

The dataset was originally created by IBM data scientists as a synthetic but realistic representation of human resources data. While it does not reflect a real organisation, it is widely used in HR analytics research and education due to its structural richness and diversity of variables.

It includes both categorical variables such as Department, JobRole, MaritalStatus and OverTime, and continuous variables such as MonthlyIncome, YearsAtCompany, and DistanceFromHome.

The target variable — Attrition — is binary, recording whether an employee left the organisation (Yes) or remained (No). Of the 1,470 employees, 237 (16.1%) are recorded as having left, making this a moderately imbalanced dataset that reflects realistic organisational turnover rates.

Methodology

Data Import and Preparation

The dataset was downloaded from Kaggle as a CSV file and imported into MySQL Workbench using the Table Data Import Wizard.

Prior to analysis, the data was inspected for missing values, duplicate records, and formatting inconsistencies in MS Excel.

No significant data quality issues were identified — the dataset was clean and required no imputation or row removal. Column names were verified against the Kaggle data dictionary to confirm accurate interpretation of coded variables, particularly satisfaction and balance scores which use numeric scales of 1 to 4.

SQL Tool Selection

All queries were written and executed in MySQL Workbench 8.0. MySQL was selected for its widespread use in industry, compatibility with the CSV import format, and robust support for the SQL techniques required by this analysis.

Visualisations were created in Microsoft Excel and Tableau using query result exports.

*AI Reference: Claude - Sonnet 4.6 (Anthropic) and ChatGPT were used as an AI learning aid for SQL syntax debugging and proof-reading and refining report content.

SQL Technique Rationale

SQL techniques were deliberately selected based on the analytical requirements of each query to ensure clarity, efficiency, and business interpretability.

CASE WHEN for Bucketing – Continuous variables such as YearsAtCompany, DistanceFromHome, and YearsSinceLastPromotion were bucketed into meaningful categorical ranges using CASE WHEN expressions. This approach was preferred over WHERE clauses because it allows all records to be retained in a single query while assigning each row to exactly one category. Bucketing transforms raw numeric data into business-readable segments — for example, converting YearsAtCompany into tenure phases such as New Hire, Early Stage and Long Term — making results directly interpretable by non-technical stakeholders.

GROUP BY for Aggregation - GROUP BY was used extensively to aggregate employee records into meaningful analytical categories such as tenure groups, departments, education levels, and promotion ranges. This enabled calculation of summary metrics including employee counts, attrition counts, and attrition rates for each segment. Grouping was essential for identifying patterns and comparing attrition behavior across different workforce categories.

ORDER BY for Trend Interpretation – ORDER BY was applied to organize query outputs in a logical and interpretable sequence. For example, tenure and promotion ranges were sorted chronologically rather than alphabetically to preserve business meaning and improve readability of trends in both SQL outputs and visualizations.

Common Table Expressions (CTEs) – CTEs were used in queries requiring multi-step logic, such as the composite risk scoring analysis and the stock option attrition query. A CTE separates each logical step into a named, readable block — the first CTE calculates a derived value (such as a risk score or total employee count), and subsequent CTEs or the final SELECT reference that result. This approach was chosen over nested subqueries because it improves readability, simplifies debugging, and makes the analytical logic auditable — important when presenting SQL to non-technical stakeholders who may review the methodology.

LEFT JOIN – Used in the stock option attrition query to ensure all stock option levels appeared in results regardless of whether attrition occurred. An INNER JOIN would have silently excluded groups with zero attrition, producing misleading results.

Conditional COUNT and AVG – Rather than filtering data with WHERE clauses and running separate queries per group, conditional aggregation using COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) was used to calculate attrition counts and rates within a single query. This produces more compact, comparable output and reduces the risk of inconsistency between separately run queries.

Data Visualization Tools

Data visualization tools - *Microsoft Excel* and *Tableau* were used to transform raw analytical outputs into clear and interpretable business insights.

Excel was primarily used for initial data organization, pivot table analysis, and creation of foundational charts such as bar charts, combo charts etc. due to its flexibility and accessibility for quick exploratory analysis.

Tableau was also used to build more interactive and visually structured dashboards that enabled clearer analysis of compensation factors across multiple categories.

Visualization using these tools improved the interpretability of findings, supported data-driven storytelling, and enhanced communication of analytical insights to non-technical stakeholders.

Limitations

The following limitations should be considered when interpreting the findings of this analysis.

Synthetic Dataset — The IBM HR Analytics dataset is synthetically generated and does not represent a real organisation. While it reflects realistic HR patterns and is structurally appropriate for SQL analysis, findings should be treated as directional rather than definitive. Causal claims cannot be made from this data — correlations observed between variables such as training frequency and attrition, or promotion recency and turnover, do not confirm that one factor causes the other.

Binary Attrition Variable — The dataset records only whether an employee left or stayed, with no information on the reason for departure — voluntary resignation, redundancy, retirement, and performance-related exits are all recorded identically. This limits the precision of recommendations, as different exit types warrant different interventions.

Scale Limitations — Several analytical subgroups contain relatively small employee counts. For example, the High Gap career segment contains only 75 employees, and the HR department contains only 63. Attrition rates calculated from small samples are more sensitive to individual cases and should be interpreted with caution.

No Longitudinal Data — The dataset represents a single point in time snapshot. It does not capture how attrition patterns change over years, whether HR interventions have previously been attempted, or how external economic conditions may have influenced turnover.

Survey-Based Variables — Satisfaction and work-life balance scores are self-reported on a 1 to 4 scale. These are subject to response bias and may not fully capture the complexity of employee sentiment. The scale labels — Low, Medium, High, Very High — are interpreted consistently throughout this analysis but their precise meaning varies by individual respondent.

Core Analysis - Who is leaving, from where, why?

To understand the impact of employee attrition on the organization we categorized the queries into four sections based on the following domains and target stakeholders. In the last section of holistic analysis, we took one advanced query each to do deeper analysis.

1. Attrition drivers (Parul)
    - Overall attrition company wide - HR Director
    - Attrition rate department wise - HR Director
    - Attrition by Job Role and Job level - Dept Manager
    - Attrition by Business travel frequency - COO
    - Attrition by marital status and gender - DEI Lead

2. Satisfaction & Engagement (Sangita)
    - Average Satisfaction Score vs Attrition Status - HR Director
    - Work-life balance score + Overtime & Attrition - COO
    - Work-life balance + No Overtime & Attrition - COO

3. Compensation (Adonay)
    - Average Monthly Income by Job Role and Level - CFO
    - Gender Pay Gap by Dept - DEI Lead
    - Salary Hike vs Performance Rating Correlation - CFO
    - Income distribution among attrition vs retained employees - HR Director
    - Stock Option level vs attrition - CFO

4. Career & Tenure (Divya)
    - Years since last promotion vs attrition - HR Director
    - Years with current manager vs attrition - Dept Manager
    - Career Progression: Total working years vs Current Role Years - COO
    - Training Investment vs Attrition - L&D Manager
    - Attrition Risk at Key tenure milestones - HR Director
    -
5. Holistic Analysis - Composite (**One query each)
    - Distance from home vs attrition by role - COO
    - Multi-company experience vs current loyalty - Talent Acquisition
    - Full Attrition Profile - CEO/Board of Directors
    - High Risk Employee Segments (composite score) - CEO Board

Business Recommendations:

We then mapped the queries to the stakeholder and segment-wise “target business suggestions” :

1. HR Director — attrition rates, satisfaction scores, high-risk employee flagging, tenure milestones
2. CFO — compensation equity, salary hike vs performance, stock options as retention lever
3. COO — overtime policy, business travel, commute distance, work-life balance
4. DEI Lead — gender pay gap, attrition by marital status/gender, demographic equity
5. Dept. Managers — role-level attrition, job involvement, manager relationship stability
6. L&D / Talent Acquisition — training ROI, education-role alignment, multi-company job-hoppers
7. CEO / Board — composite risk score, full leaver vs stayer profile summary

SECTION -1

ATTRITION DRIVERS

Section Owner - Parul

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Introduction

This section analyzes the key factors influencing employee attrition within the organization. Using SQL-based aggregation, demographic segmentation, and data visualization techniques, the analysis examines how attrition varies across managerial tenure, promotion history, demographics, education, and organizational characteristics. The objective is to identify patterns associated with higher employee turnover and determine which workforce segments demonstrate elevated attrition risk.

By translating raw employee data into interpretable business insights, this section supports evidence-based recommendations aimed at improving employee retention, strengthening workforce stability, and enhancing managerial effectiveness.

Query 1: Overall company wide attrition rate

Stakeholders: HR Director

Research Question: What is the company-wide attrition rate? This query aimed at understanding the overall attrition rate before delving deeper into the drivers of attrition.

SQL Query:

```
SELECT
  COUNT(*) AS Total_Employees,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
  COUNT(CASE WHEN Attrition = 'No' THEN 1 END) AS Retained_Count,
  ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm;
```

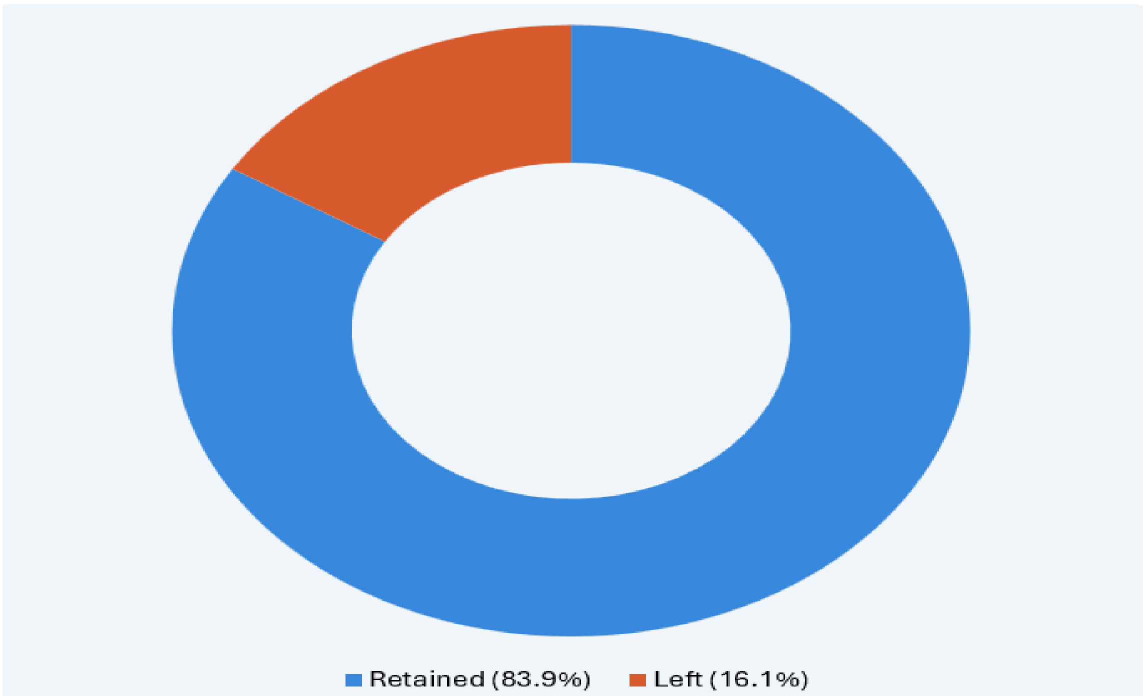
Explanation:

The query used Count and Case When to calculate the attrition rate within the company.

Results Table

Total_Employees	Attrition_Count	Retained_Count	Attrition_Rate
1470	237	1233	16.1

Chart:



Insights: This query gives a company-wide attrition summary. It counts total employees, how many left, how many stayed, and calculates the overall turnover rate as a percentage from the IBM table. The donut chart shows the overall attrition rate and employees retained as a percentage.

Business Recommendations: Investigate root causes of the 16.1% attrition rate as it signals a systemic retention problem requiring company-wide action.

Query 2: Overall attrition rate department wise

Stakeholders: HR Director

Research Question: Does attrition rate vary across departments?

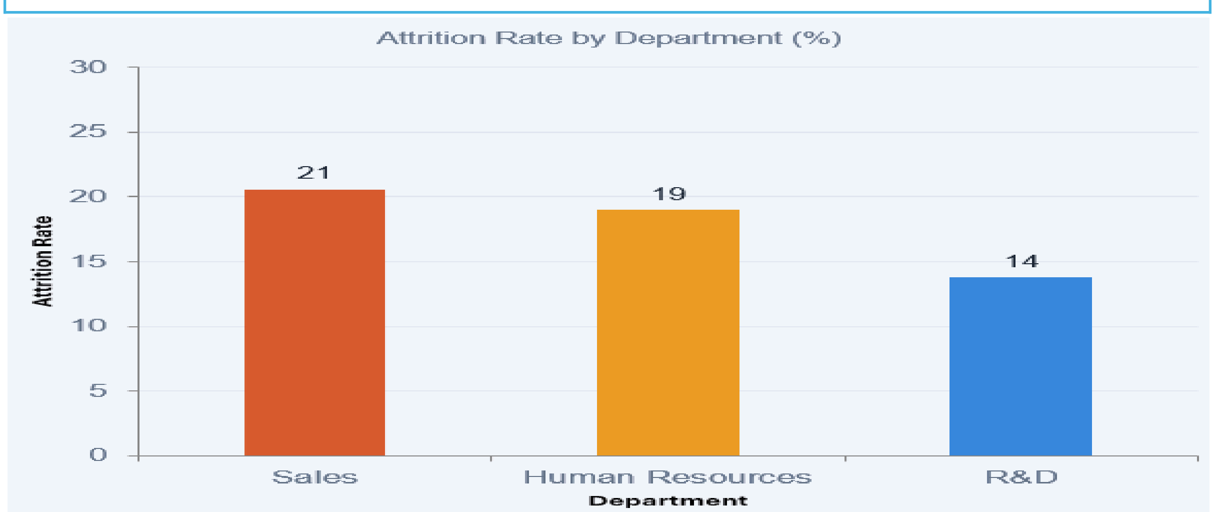
SQL Query:

```
SELECT
   Department,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
   COUNT(*) AS Total_Employees,
   ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY Department
ORDER BY Attrition_Rate DESC;
```

Explanation: This query groups employees by Department and calculates the attrition count, total headcount, and attrition rate for each one, ordered from highest to lowest rate. It gives a cleaner, department-level view of retention, useful for identifying which business units need the most attention without the noise of individual job roles.

Results Table & Chart:

Department	Attrition_Count	Total_Employees	Attrition_Rate
Sales	92	446	20.6
Human Resources	12	63	19
Research & Development	133	961	13.8



* Figure: Attrition Rate by Department

Insights: From the bar chart, we can see that Sales has the highest attrition rate at 20.6%, followed closely by Human Resources at 19.0%, while Research & Development is the most stable at 13.8% despite having the largest workforce of 961 employees. This confirms that Sales is the highest-risk department overall, and even though R&D has the most departures in raw numbers (133), its large headcount keeps the rate relatively lower.

Business Recommendations: Prioritize retention efforts for Sales Representatives and Laboratory Technicians through better compensation, workload balance, and career growth paths.

Query 3: Attrition by Department and Job Role

Stakeholder: HR Director / Department Heads

Research Question: Does attrition vary across different job roles and departments?

SQL Query:

```
SELECT
  Department,
  JobRole,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
  COUNT(*) AS Total_Employees,
  ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY Department, JobRole
ORDER BY Attrition_Rate DESC;
```

Explanation:

This query breaks down attrition at a more granular level by using GROUP BY to group employees by both Department and Job Role. It calculates the attrition count and rate for each combination, then orders the results from highest to lowest attrition rate, making it easy to pinpoint which specific roles and departments are losing the most employees.

Results Table & Chart:

Department	JobRole	Attrition_Count	Total_Employees	Attrition_Rate

Human Resources	Human Resources	12	52	23.1
Human Resources	Manager	0	11	0
Research & Development	Laboratory Technician	62	259	23.9
Research & Development	Research Scientist	47	292	16.1
Research & Development	Manufacturing Director	10	145	6.9
Research & Development	Healthcare Representative	9	131	6.9
Research & Development	Manager	3	54	5.6
Research & Development	Research Director	2	80	2.5
Sales	Sales Representative	33	83	39.8
Sales	Sales Executive	57	326	17.5
Sales	Manager	2	37	5.4

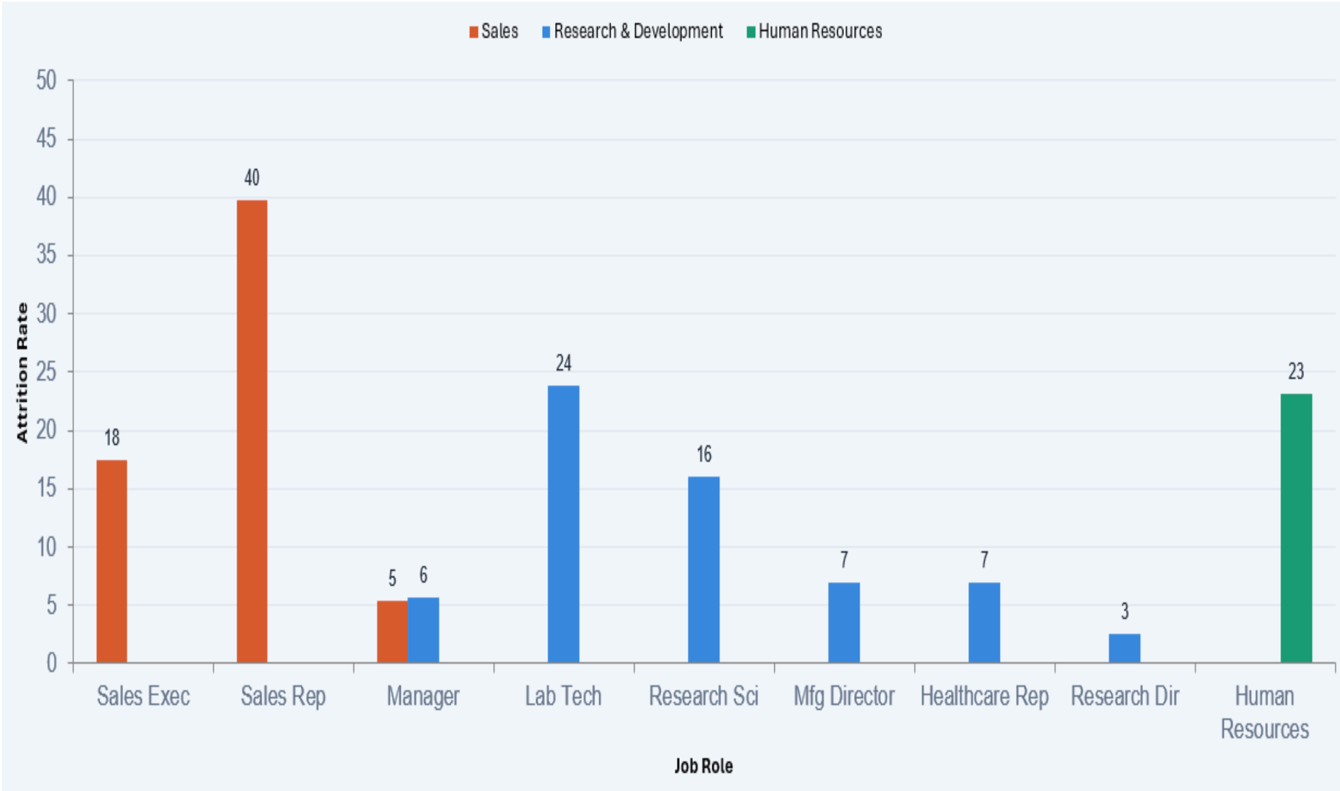


*Figure: Attrition by Department & Job Role

Insights:

The Bar chart depicts attrition rate across all departments by job roles. It is evident that the Sales Representative has the highest Attrition Rate at 39.8% with nearly 4 in 10 employees leaving, which is a major retention concern. Laboratory Technicians (23.9%) and HR staff (23.1%) also show notably high turnover. Sales Executives (17.5%) and Research Scientists (16.1%) sit in the middle, still significant given their large headcounts of 326 and 292 employees respectively.

Managers across all departments show very low or zero attrition, suggesting senior roles have stronger retention. Research Directors at 2.5% are the most stable role in the dataset. Overall, Customer-facing and technical entry-level roles struggle the most with retention, while managerial and senior research roles are the most stable. The Sales department as a whole appears to be the highest-risk area for attrition.

Business Recommendation: Focus on the Sales department with targeted engagement programs and incentive restructuring.

Query 4: Travel Impact on Attrition

Stakeholders: HR & Operations Management

Research Question: Do employees with business travel as a part of their job role leave more?

SQL Query:

```
SELECT
   BusinessTravel,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
   COUNT(*) AS Total_Employees,
    ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*),
1) AS Attrition_Rate
FROM ibm
GROUP BY BusinessTravel
ORDER BY Attrition_Rate DESC;
```

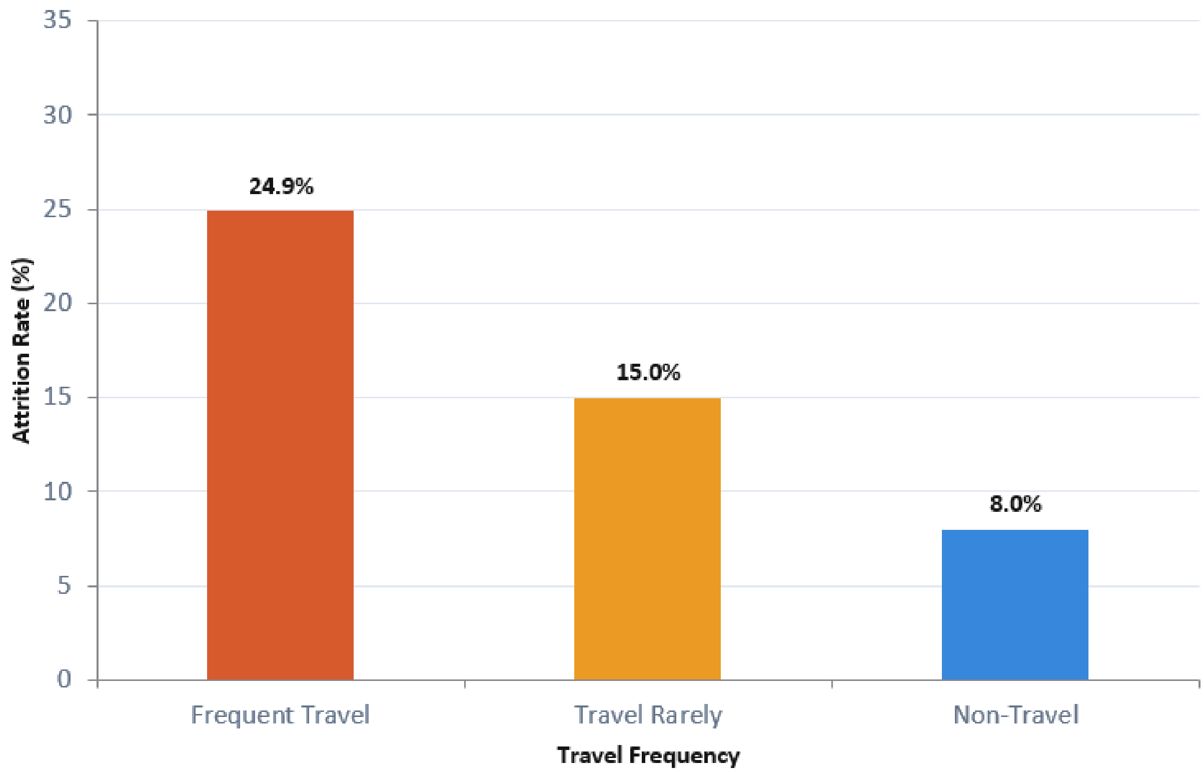
Explanation:

This query groups employees by their travel frequency and calculates the attrition count, total headcount, and attrition rate for each group, ordered from highest to lowest. It helps identify whether how often an employee travels has any relationship with their likelihood of leaving the company.

Result Table:

BusinessTravel	Attrition_Count	Total_Employees	Attrition_Rate
Travel_Frequently	69	277	24.9
Travel_Rarely	156	1043	15
Non-Travel	12	150	8

Chart:



Insights

Travel frequency has a clear and direct relationship with attrition, the more an employee travels, the more likely they are to leave. Frequent travelers have the highest attrition at 24.9%, nearly 1 in 4 employees, suggesting travel burnout is a real factor. Rare travelers sit at 15%, which is moderate but still significant given they make up the largest group at 1,043 employees. Non-travelers are the most stable at just 8%, less than half the rate of frequent travelers.

Business Recommendations:

Reduce unnecessary frequent travel, offer travel allowances or remote work alternatives to ease burnout for high-travel roles

Query 5: Attrition by Age

Stakeholders: HR & Talent Acquisition Team

The query aims to study attrition in employees across different age brackets.

Research Question: Does age have any impact on employee attrition? Are employees more loyal to a company at later stages in life?

SQL Query:

```
SELECT
CASE
    WHEN Age < 25 THEN 'Under 25'
    WHEN Age BETWEEN 25 AND 34 THEN '25-34'
    WHEN Age BETWEEN 35 AND 44 THEN '35-44'
    WHEN Age BETWEEN 45 AND 54 THEN '45-54'
    ELSE '55+'
  END AS Age_Group,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
  COUNT(*) AS Total_Employees,
  ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY Age_Group
ORDER BY Attrition_Rate DESC;
```

Explanation:

This query buckets employees into age ranges using a CASE WHEN statement, then calculates attrition for each group.

Result Table & Chart:

Age_Group	Attrition_Count	Total_Employees	Attrition_Rate
Under 25	38	97	39.2
25-34	112	554	20.2
55+	11	69	15.9
45-54	25	245	10.2
35-44	51	505	10.1

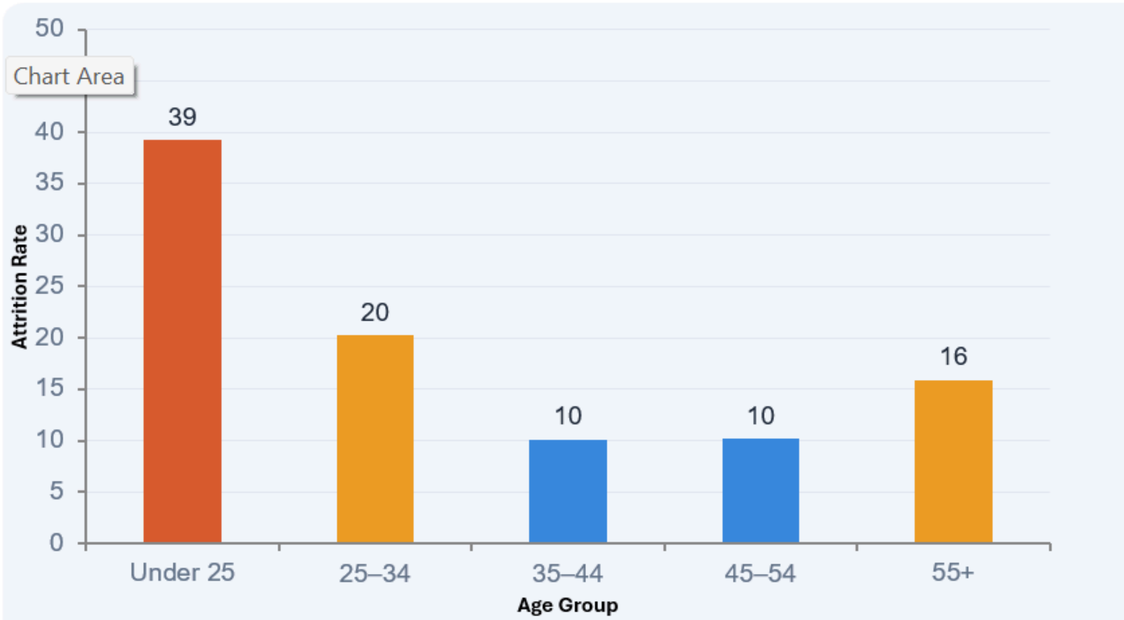


*Figure: Attrition Rate by Group

Insights:

Age has a strong inverse relationship with attrition. Younger employees leave at significantly higher rates. Overall, The company's biggest retention challenge is with young employees under 35. Investing in early-career development programs, mentorship, and competitive growth opportunities could significantly reduce attrition in these age groups.

- Under 25 has a striking attrition rate of 39%, the highest by far, suggesting early-career employees are the hardest to retain.
- 25–34 drops to 20%, still elevated, likely reflecting employees who are actively exploring career options.
- 35–44 and 45–54 are the most stable groups, both at just 10%, indicating mid-career employees are far more settled.
- 55+ rises slightly to 16%, possibly reflecting early retirements or end-of-career transitions.

Business Recommendation: Launch mentorship and early career development programs targeting employees under 35 to improve engagement and long-term commitment

Query 6 : Attrition Rate by Marital Status

Stakeholders: HR & Diversity and Inclusion Team

Research Question: Does marital status impact attrition among employees?

SQL Query:

```
SELECT
   Gender,
   MaritalStatus,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
   COUNT(*) AS Total_Employees,
   ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY Gender, MaritalStatus
ORDER BY Attrition_Rate DESC;
```

Explanation:

This query was aimed at analyzing the role of marital status on employee attrition. It helps uncover whether personal life circumstances combined with gender have any influence on an employee's likelihood of leaving.

This query groups employees by both Gender and Marital Status together, calculating attrition count, total headcount, and rate for each combination.

Results Table:

Gender	MaritalStatus	Attrition_Count	Total_Employees	Attrition_Rate
Male	Single	73	271	26.9
Female	Single	47	199	23.6
Male	Married	53	401	13.2
Female	Married	31	272	11.4
Male	Divorced	24	210	11.4
Female	Divorced	9	117	7.7

Chart:

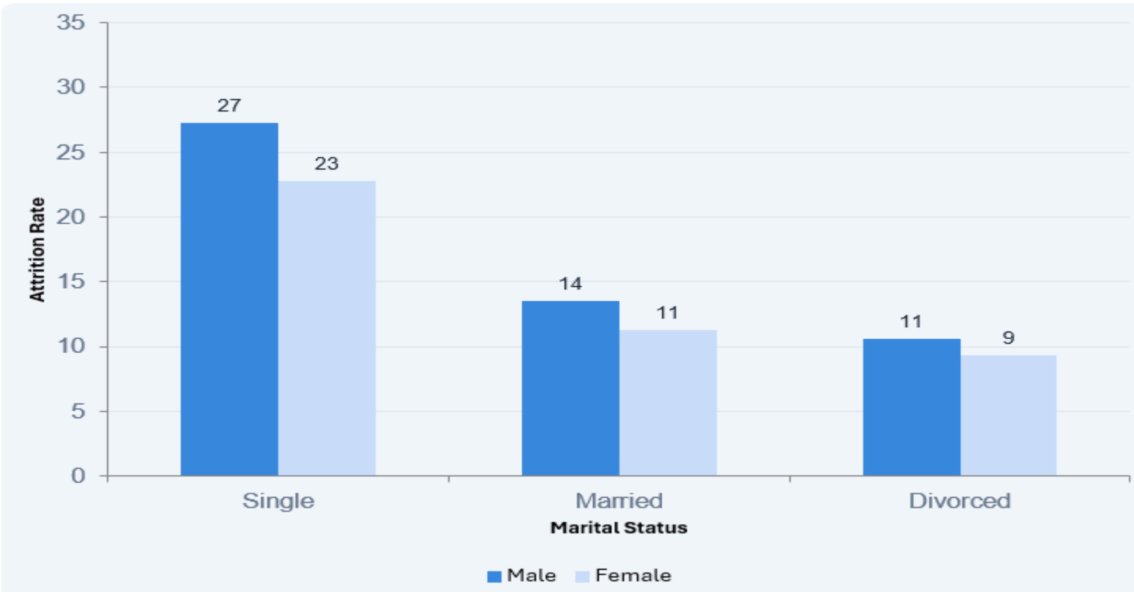


*Figure: Attrition Rate by Marital Status

Insights:

Marital status has a stronger influence on attrition than gender, though both play a role. Overall, Single employees, regardless of gender, are the highest flight risk. Across all marital status groups, males consistently leave at slightly higher rates than females. Retention strategies targeting younger, single employees could have the greatest impact.

- Single employees have the highest attrition across both genders, 26.9% for single males and 23.6% for single females, likely because they have fewer financial obligations and more flexibility to job-hop.
- Married employees sit in the middle at 13.2% (male) and 11.4% (female), showing that marriage correlates with greater stability.
- Divorced employees have the lowest attrition, 11.4% (male) and just 7.7% (female), possibly reflecting a greater need for job security and income stability.

Recommendations: Design retention programs for single employees such as social engagement initiatives, flexible benefits, and clearer career progression for better engagement and sense of a robust professional growth.

Query 7: Attrition rate by number of prior employers

Stakeholder: Recruitment / Hiring Managers

Research Question: Does the number of prior employers impact employee attrition? Can no of companies previous to joining current one determine if an employee is at higher attrition risk?

SQL Query:

```
SELECT
   NumCompaniesWorked,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
   COUNT(*) AS Total_Employees,
   ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY NumCompaniesWorked
ORDER BY NumCompaniesWorked;
```

Explanation:

The query aims to analyse the impact of job history on employee attrition

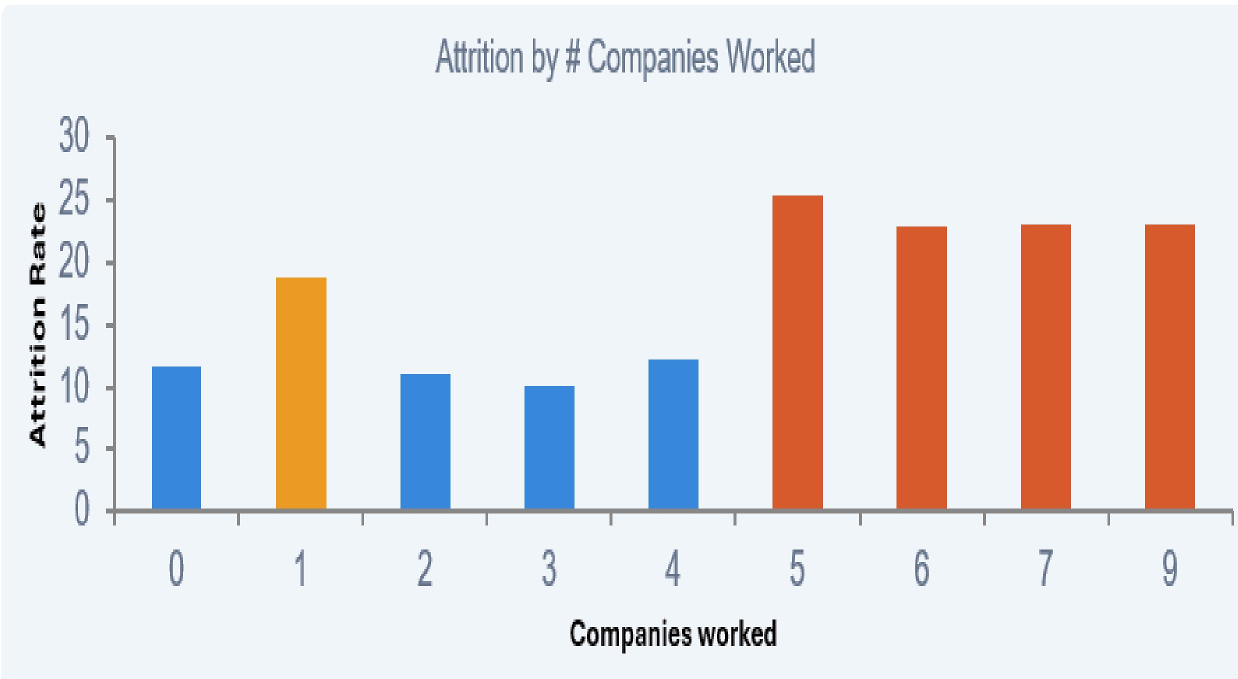
It groups employees by how many companies they've worked at previously and calculates attrition for each value. It helps determine whether prior job-switching behavior predicts future attrition.

Results Table:

NumCompaniesWorked	Attrition_Count	Total_Employees	Attrition_Rate
0	23	197	11.7
1	98	521	18.8
2	16	146	11
3	16	159	10.1
4	17	139	12.2
5	16	63	25.4
6	16	70	22.9

7	17	74	23
8	6	49	12.2
9	12	52	23.1

Chart:



Insight:

There is no perfectly linear pattern, but employees who have worked at 5, 6, 7, or 9 companies show notably high attrition rates of 25.4%, 22.9%, 23%, and 23.1% respectively, suggesting that serial job-switchers tend to continue that behavior. Employees who worked at 2 - 4 companies are the most stable, hovering around 10 -12%, while those who only ever worked at 1 company (18.8%) leave more than expected, possibly due to lack of perspective on what other employers offer.

Business Recommendation: Flag employees with 5+ prior companies for early engagement conversations and personalized development plans.

Query 9 -

Stakeholder: Recruitment / HR Strategy Team

Research Question: Do job hoppers have a higher attrition rate?

SQL Query:

```
SELECT
   CASE
      WHEN NumCompaniesWorked >= 4 AND YearsAtCompany <= 3 THEN 'Job Hopper'
      WHEN NumCompaniesWorked <= 2 AND YearsAtCompany >= 5 THEN 'Loyal
Employee'
      ELSE 'In Between'
   END AS Employee_Type,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS Attrition_Count,
   COUNT(*) AS Total_Employees,
   ROUND(COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*), 1) AS
Attrition_Rate
FROM ibm
GROUP BY Employee_Type
ORDER BY Attrition_Rate DESC;
```

Explanation:

This advanced query was included as supplementary analysis to support the Talent Acquisition stakeholder perspective.

This query classifies employees into three types using two conditions combined, prior company count and years at the current company. A Job Hopper has worked at 4+ companies and has been here 3 years or less, a Loyal Employee has worked at 2 or fewer companies and has been here 5+ years, and everyone else falls In Between. It then measures attrition for each type.

*<u>*This was not included in the presentation and hence, does not include a graphical presentation of the results*</u>.

Result Table:

Employee_Type	Attrition_Count	Total_Employees	Attrition_Rate
Job Hopper	41	180	22.8
In Between	139	717	19.4

Loyal Employee	57	573	9.9

Insights:

Overall Insight Employment history is a meaningful attrition signal. Employees with a pattern of frequent company changes are twice as likely to leave as loyal, long-tenured employees. This could inform hiring decisions or prompt earlier engagement efforts for employees flagged as high flight risks.

Job Hoppers have the highest attrition at 22.8%, their history of frequent switching carries over into their current role.

In Between employees sit at 19.4%, still a meaningful risk given they make up the largest group of 717 employees.

Loyal Employees have by far the lowest attrition at just 9.9%, less than half that of job hoppers, confirming that past loyalty is a strong predictor of future retention.

Business Recommendation: During hiring, prioritize candidates with stable employment histories and invest in retaining identified loyal employees.

SECTION -2

SATISFACTION
&
ENGAGEMENT

Section Owner:
Sangita

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Introduction

This section focuses on the Satisfaction & Engagement theme, examining how job satisfaction levels, work-life balance scores, and overtime requirements relate to voluntary employee turnover. A third exploratory query investigated commute distance by job role; while it produced an interesting SQL structure, the results were inconsistent across roles and commute distance did not emerge as a strong standalone predictor of attrition.

Query 1: Job Satisfaction VS Attrition

Stakeholder: HR Director - Retention Strategy & Policy Design

Research Question:

Does job satisfaction level predict attrition rate? Can satisfaction scores serve as an early warning signal for HR directors to identify at-risk employee segments before turnover occurs?

SQL Query

```
 SELECT
   CASE
      WHEN JobSatisfaction = 1 THEN 'Low'
      WHEN JobSatisfaction = 2 THEN 'Moderate'
      WHEN JobSatisfaction = 3 THEN 'Good'
      WHEN JobSatisfaction = 4 THEN 'Very Good'
   END AS JobSatisfaction,
   ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN 1.0 ELSE 0 END) * 100, 1)
      AS Attrition_Rate_in_Percentage
FROM HR
GROUP BY JobSatisfaction
ORDER BY
   CASE
      WHEN JobSatisfaction = 1 THEN 1
      WHEN JobSatisfaction = 2 THEN 2
      WHEN JobSatisfaction = 3 THEN 3
      WHEN JobSatisfaction = 4 THEN 4
   END;
```

Query explanation: The query uses a CASE WHEN expression to convert numeric satisfaction codes (1 to 4) into human-readable labels. The attrition rate is computed using AVG on a conditional CASE WHEN that returns 1.0 for employees who left and 0 for those who stayed, then multiplied by 100. A second CASE WHEN in the ORDER BY clause enforces

Low to Very Good ordering rather than alphabetical sort, which would otherwise misrepresent the trend.

Result Table & Chart

Job Satisfaction Level	Attrition Rate (%)
Low	22.8
Moderate	16.4
Good	16.5
Very Good	11.3

Insights: Result shows that Employees with Low job satisfaction leave at 22.8% nearly double the 11.3% rate for those with Very Good satisfaction, confirming a meaningful inverse relationship between satisfaction and attrition. The relationship is not perfectly linear: Moderate and Good satisfaction groups show near-identical attrition rates (16.4% and 16.5%), suggesting diminishing marginal returns between middle satisfaction levels. Even employees reporting Very Good satisfaction leave at 11.3%, indicating that job satisfaction alone is not sufficient to fully retain employees, other factors are also at play.

Business Recommendations: The Key stakeholder for this would be HR Directors. From this we suggest implementing regular team-level satisfaction pulse surveys to track trends over time. Declining satisfaction scores particularly toward the Low or Moderate range should trigger proactive check-ins and targeted retention actions before employees reach the point of departure. We believe that satisfaction data should be treated as a leading indicator of upcoming turnover risk.

Query 2: Work Life Balance + Overtime VS Attrition

Stakeholder: COO - Workload Planning & Burnout Prevention

Research Question:

Does working overtime significantly compound attrition risk beyond what work-life balance score alone predicts? And does this effect persist across all balance levels, or only at the extremes?

SQL Query

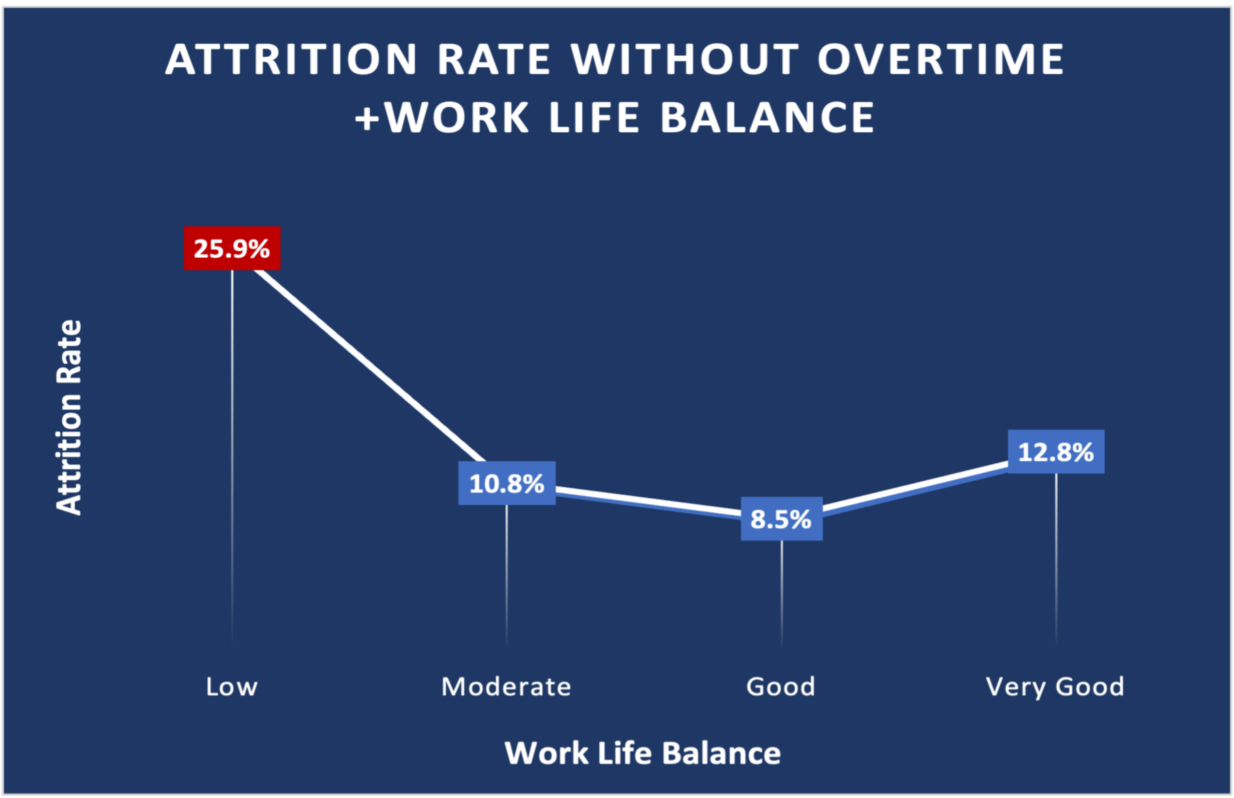
```
SELECT
   CASE WorkLifeBalance
      WHEN 1 THEN 'Low'
      WHEN 2 THEN 'Moderate'
      WHEN 3 THEN 'Good'
      WHEN 4 THEN 'Very Good'
   END AS WorkLifeBalance,
   OverTime,
   ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
      AS Attrition_Rate_in_Percentage
FROM HR
GROUP BY WorkLifeBalance, OverTime
ORDER BY WorkLifeBalance, OverTime;
```

Explanation: This query extends the first analysis by introducing OverTime as a second grouping dimension. The CASE expression on WorkLifeBalance uses the shorthand CASE column WHEN syntax. Attrition rate is calculated using SUM divided by COUNT rather than AVG, producing the same result but making the logic more explicit. GROUP BY WorkLifeBalance, OverTime generates eight output rows one per combination enabling a direct side-by-side comparison of overtime versus non-overtime groups at each balance level.

Result Tables & Charts

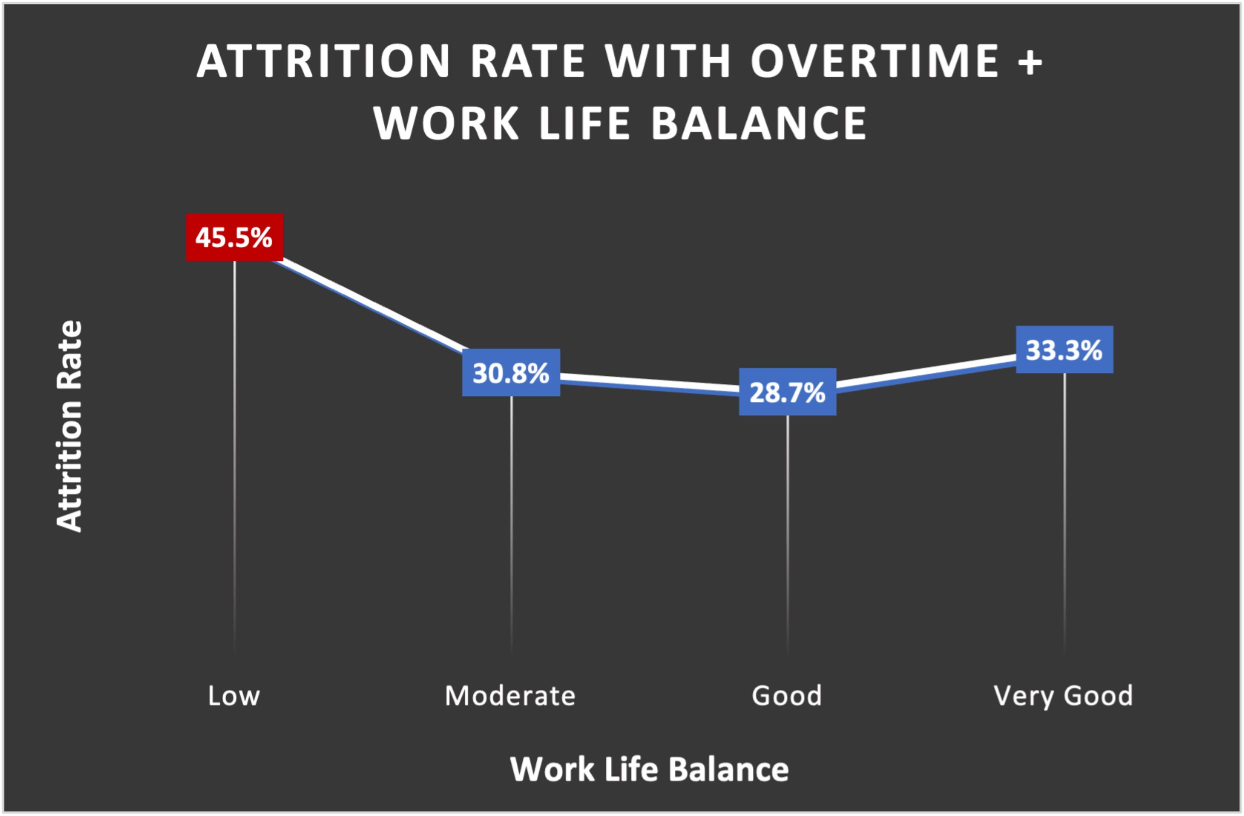
Without Overtime

Work-Life Balance	Attrition Rate (%)
Low	25.9
Moderate	10.8
Good	8.5
Very Good	12.8



With Overtime

Work-Life Balance	Attrition Rate (%)
Low	45.5
Moderate	30.8
Good	28.7
Very Good	33.3



Insights: The table results show that overtime nearly doubles attrition risk at every work-life balance (WLB) level. For example, employees with Good WLB leave at 8.5% without overtime, but 28.7% with overtime, a 3.4x increase from the same satisfaction baseline.

Without overtime, attrition trends generally downward as WLB improves and ranges from 8.5% to 25.9% difficult at the low end, but manageable. With overtime, all groups remain critically elevated between 28.7% and 45.5%. Even employees with Very Good work-life balance who work overtime leave at 33.3%, demonstrating that self-reported balance does not protect against attrition when overtime is required, the structural demand overrides the perceived balance.

Business Recommendations: We suggest that the COO should treat overtime reduction as the single highest ROI lever for immediate attrition impact. Capping or eliminating sustained overtime especially for teams already reporting low work-life balance, which could reduce attrition rates by up to half in the highest-risk groups. We suggest that workload distribution, headcount planning, and project scheduling should be evaluated before approving ongoing overtime commitments.

Query 3: Distance from Home VS Attrition by Job Role

Stakeholder: COO - Flexible Work & Commute Policy

Research Question

Does commute distance predict attrition, and does its impact vary meaningfully across different job roles? This query was included as a supplementary exploratory analysis.

SQL Query

```
SELECT
   JobRole,
   CASE
      WHEN DistanceFromHome <= 5 THEN 'Near'
      WHEN DistanceFromHome <= 15 THEN 'Moderate'
      ELSE 'Far'
   END AS Distance_Group,
    ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1)
AS Attrition_Rate_in_Percentage
FROM HR
GROUP BY JobRole, Distance_Group
ORDER BY JobRole, Distance_Group;
```

Explanation: *This is the most structurally complex query of the three.*

It uses a nested CASE WHEN expression to bucket the continuous DistanceFromHome variable into three categorical distance bands. Grouping by both JobRole and Distance_Group produces a matrix output of 27 rows across 9 roles and 3 distance groups, enabling cross-role comparison of the commute-attrition relationship. The WHERE-less design ensures all role-distance combinations are represented, including those with low attrition.

Result Table

Job Role	Distance Group	Attrition Rate (%)
Human Resources	Near (≤5 mi)	8.0
Human Resources	Moderate (6–15 mi)	25.0
Human Resources	Far (>15 mi)	54.5
Sales Representative	Near (≤5 mi)	41.7
Sales Representative	Moderate (6–15 mi)	37.5
Sales Representative	Far (>15 mi)	40.0

Laboratory Technician	Near (≤5 mi)	22.3
Laboratory Technician	Moderate (6–15 mi)	26.3
Laboratory Technician	Far (>15 mi)	23.0
Manager	Near (≤5 mi)	5.2
Manager	Moderate (6–15 mi)	4.0
Manager	Far (>15 mi)	5.3
Research Director	Near (≤5 mi)	2.9
Research Director	Moderate (6–15 mi)	2.9
Research Director	Far (>15 mi)	0.0

Insights: From the result table, Human Resources shows extreme distance sensitivity: attrition rises from 8% (Near) to 54.5% (Far) a 6.8x difference making it the role most impacted by commute distance in this dataset. However, Sales Representatives leave at consistently high rates regardless of distance (37-42%), indicating that commute is not the primary driver of attrition for this role group.

The overall pattern is inconsistent across roles: Laboratory Technicians show little distance effect (22-26%), while Managers and Research Directors remain low across all distances.

While the SQL for this query is the most technically sophisticated using CASE WHEN for distance bucketing across a two-dimension GROUP BY, the results are ultimately inconclusive as a standalone attrition factor. The wide variation across job roles means no single policy conclusion applies to the full workforce. This finding is itself analytically useful: it cautions against broad commute-based policies and we suggest that any flexible work interventions should be targeted by role rather than applied company-wide.

Distance does not emerge as a universal attrition predictor.

Business Recommendations: We suggest that the COO should consider role-targeted flexible or remote work options rather than blanket commute policies. For Human Resources specifically where distance shows a dramatic attrition impact offering remote or hybrid flexibility could meaningfully reduce turnover. For Sales Representatives, retention strategies should focus on other factors such as compensation, management quality, or workload, as commute distance does not appear to drive their high attrition rate.

SECTION - 3

Employee Compensation & Attrition Analysis

Prepared By:
Adonay Kfleyesus

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Introduction

This section examines how compensation structures, pay equity, and financial incentives relate to employee attrition across the organisation. Five analyses explore average income distribution by job role and level, gender pay equity across departments, the relationship between salary hikes and performance ratings, income differences between retained and attrited employees, and the impact of stock option level on turnover. Together these analyses address whether financial factors independently drive attrition or whether compensation operates alongside other retention levers. All queries were written and executed in MySQL Workbench. Charts were created in Microsoft Excel.

Query 1. Average Monthly Income by Job Role and Level

Stakeholder: CFO

Objective: This analysis is to evaluate how average monthly income varies across job roles and job levels in order to identify compensation patterns, pay gaps, and overall salary structure efficiency. We aim to determine whether pay aligns with both job level and role value, while also highlighting any inconsistencies across positions at the same level.

SQL Query

```
SELECT JobRole, JobLevel, AVG(MonthlyIncome) AS avg_income
FROM IBM
GROUP BY JobRole, JobLevel
ORDER BY JobLevel, avg_income DESC;
```

Explanation:

This query was created to better understand how salaries are distributed across different positions and employee levels within the company. By calculating the average monthly income for each role and level combination, the analysis makes it easier to see how compensation changes as employees move into higher positions. The results showed a steady increase in salary as job level increased, with leadership and management roles earning the highest incomes compared to technical and entry-level positions.

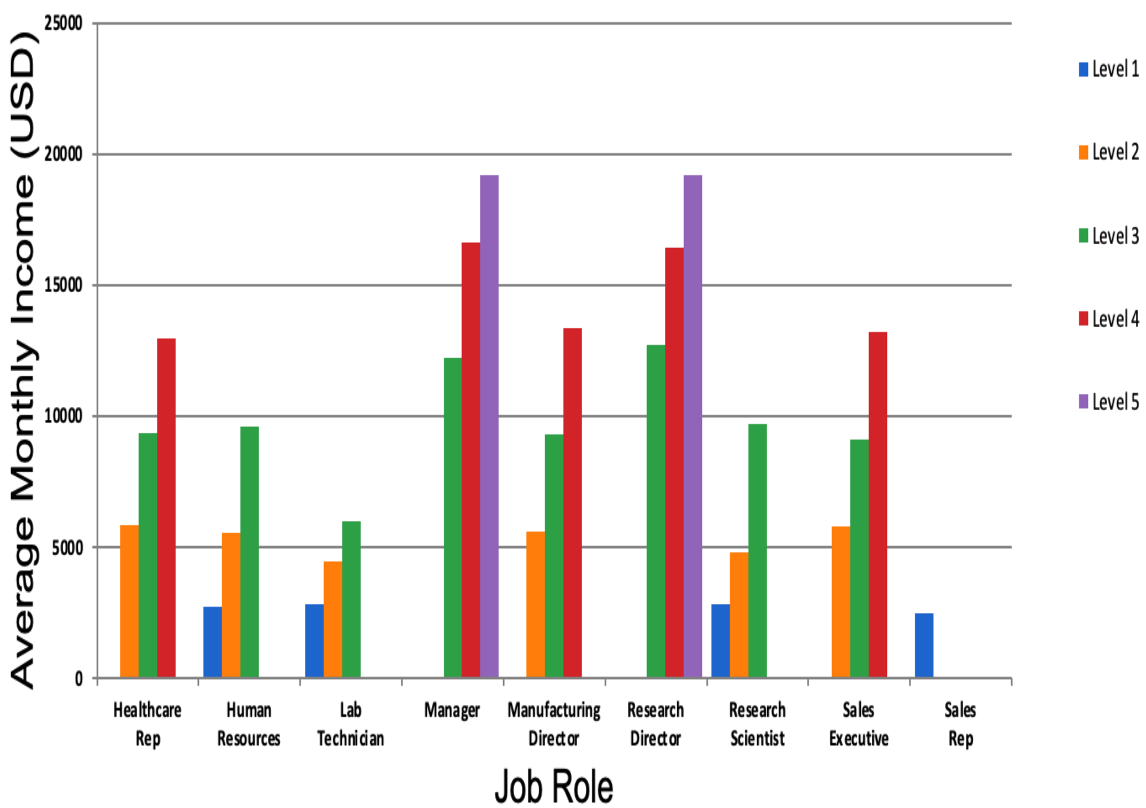
Results Table:

Average Monthly Income by Job Role & Level

Job Role	Level 1	Level 2	Level 3	Level 4	Level 5

Healthcare Representative	-	5,865	9,364	12,973	-
Human Resources	2,733	5,563	9,623	-	-
Laboratory Technician	2,855	4,456	5,998	-	-
Manager	-	-	12,233	16,613	19,184
Manufacturing Director	-	5,613	9,307	13,383	-
Research Director	-	-	12,722	16,429	19,205
Research Scientist	2,828	4,819	9,724	-	-
Sales Executive	-	5,801	9,125	13,203	-
Sales Representative	2,507	-	-	-	-

Chart:



Insights:

- Average income increases consistently as job level rises, showing that the company has a structured salary progression system tied to employee seniority and responsibility.
- Even within the same job level, managerial and director roles earn significantly higher salaries than technical or entry-level positions, highlighting clear compensation differences across roles.
- Overall, the analysis shows that employee compensation is influenced by both job level and job role, helping leadership better identify pay distribution patterns and potential pay gaps across the organization.

Business Recommendations:

- Conduct regular compensation reviews to ensure salaries remain competitive and balanced across similar job levels and roles.
- Create stronger career development and internal promotion pathways to help lower-level employees advance into higher-paying positions.

Query 2: Gender Pay Gap By Department

Research Question: Does gender pay parity across departments impact attrition?

SQL Query

```
SELECT
  Department,

  COUNT(CASE WHEN Gender = 'Female' THEN 1 END) AS female_count,
  AVG(CASE WHEN Gender = 'Female' THEN MonthlyIncome END) AS female_avg_income,

  COUNT(CASE WHEN Gender = 'Male' THEN 1 END) AS male_count,
  AVG(CASE WHEN Gender = 'Male' THEN MonthlyIncome END) AS male_avg_income,

  COUNT(*) AS total_employees,

  (
    AVG(CASE WHEN Gender = 'Female' THEN MonthlyIncome END) -
    AVG(CASE WHEN Gender = 'Male' THEN MonthlyIncome END)
```

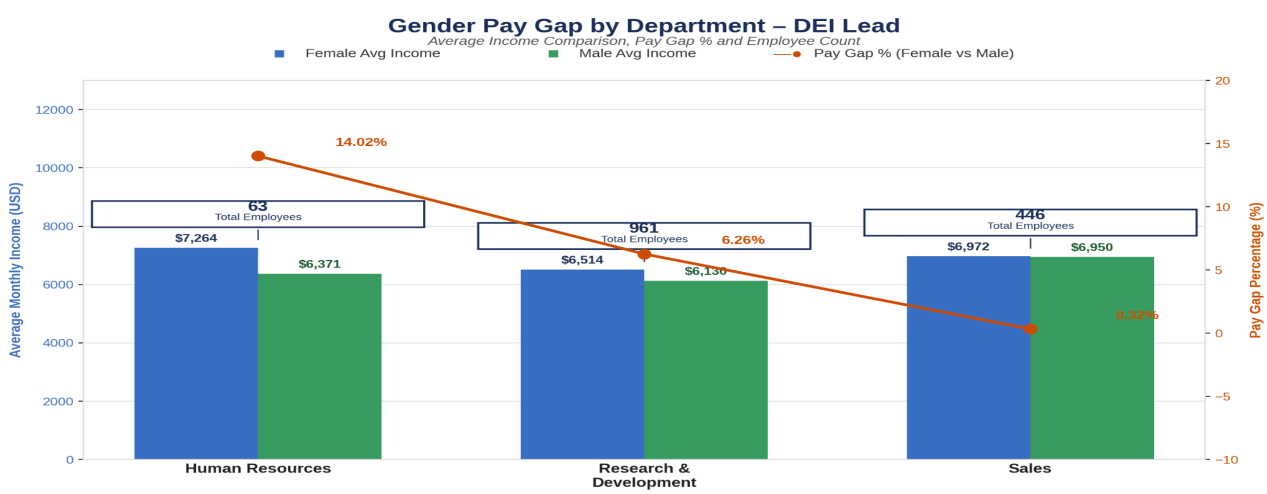
```
  ) * 100.0 /
  AVG(CASE WHEN Gender = 'Male' THEN MonthlyIncome END) AS pay_gap_percent

FROM IBM
GROUP BY Department
ORDER BY Department;
```

Explanation: This query examines salary differences between male and female employees within each department. It measures employee counts, average income by gender, and the percentage difference in pay to provide a clearer picture of compensation fairness. The findings showed that salaries between genders were generally very close across departments, with only minor differences that appeared to be influenced more by department size and workforce distribution than unequal pay practices.

Results Table & Chart:

Gender Pay Gap By Department:

Department	Female Count	Male Count	Total Employees	Female Avg	Male Avg	Pay Gap %
Human Resources	20	43	63	7,264	6,371	+14.02%
Research & Development	379	582	961	6,514	6,130	+6.26%
Sales	189	257	446	6,972	6,950	+0.32%



Department	Female Count	Male Count	Total Employees	Female Avg Income	Male Avg Income	Pay Gap %
Human Resources	20	43	63	$7,264	$6,371	14.02%
Research & Development	379	582	961	$6,514	$6,130	6.26%
Sales	189	257	446	$6,972	$6,950	0.32%

☐ Key Insight: Average salaries between male and female employees are closely aligned across all departments.
Female employees earn slightly more on average in Human Resources and R&D, while Sales shows near-perfect parity.

Insights:

The analysis shows that average salaries between male and female employees are highly comparable across departments, indicating a strong level of overall pay balance within the organization.

- Female employees earn slightly higher average salaries in certain departments, particularly in Human Resources, suggesting that salary differences are influenced more by role distribution and workforce composition than direct pay inequality.

- By including employee counts and pay gap percentages, the analysis provides additional context behind salary variations, showing that larger departments produce more stable compensation trends while smaller departments may experience greater fluctuations due to lower employee representation.

Business Recommendations:

- Regular pay equity audits to maintain fair compensation practices across all departments.
- Strengthen diversity, promotion, and leadership development initiatives to ensure equal advancement opportunities for all employees

QUERY 3: Salary Hike vs. Performance Rating:

Research Question: How does performance rating impact salary hike and in turn impact attrition?

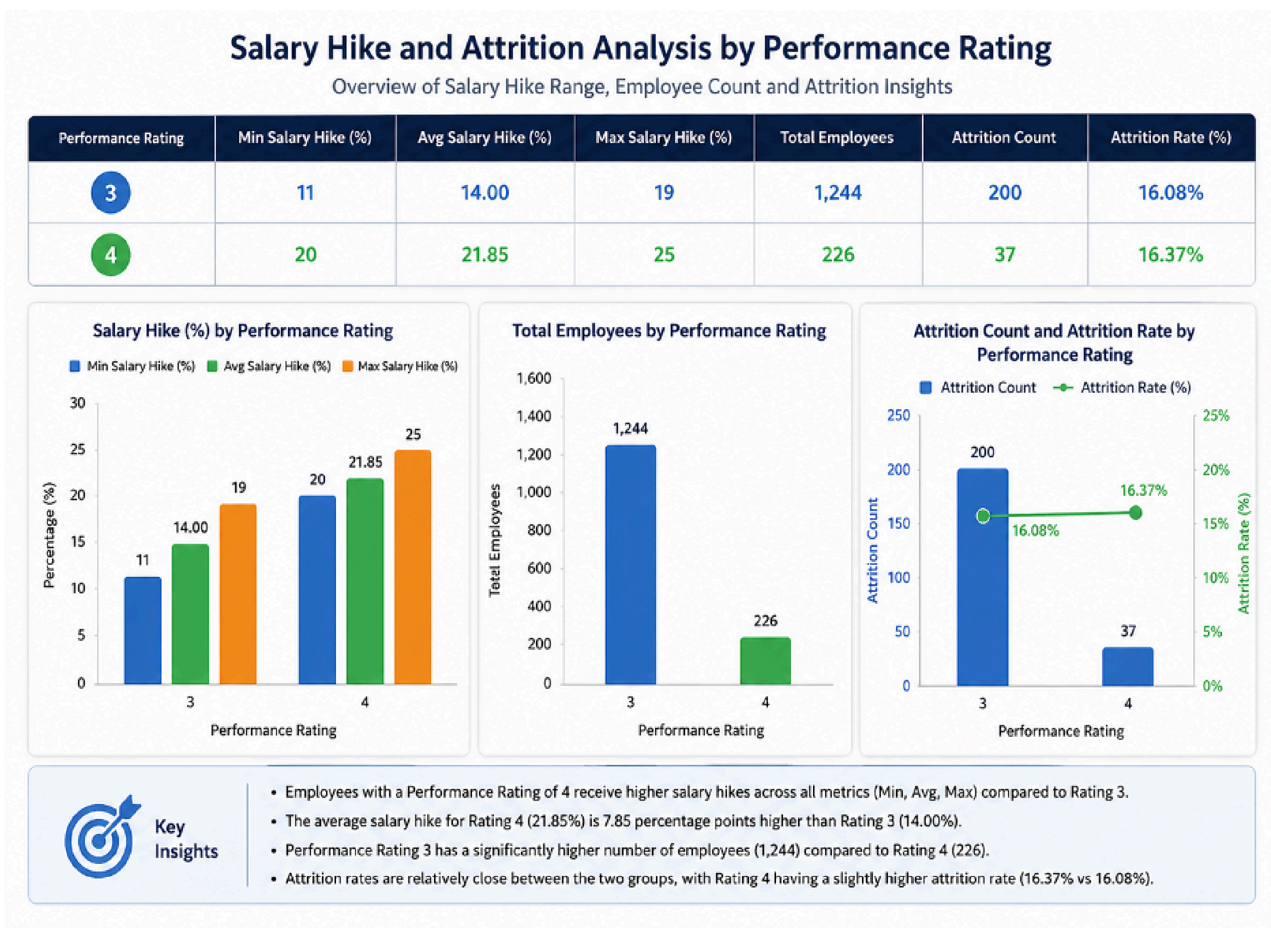
SQL Query

```
SELECT
  PerformanceRating,
      MIN(PercentSalaryHike) AS min_salary_hike,
      AVG(PercentSalaryHike) AS avg_salary_hike,
      MAX(PercentSalaryHike) AS max_salary_hike,

  COUNT(*) AS total_employees,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*) AS attrition_rate

FROM IBM
GROUP BY PerformanceRating
ORDER BY PerformanceRating;
```

Explanation: This query was designed to evaluate whether employee raises are connected to performance ratings. It compares minimum, average, and maximum salary increases across performance groups while also looking at attrition rates. The results showed that employees with stronger performance ratings received noticeably larger raises, confirming that the company rewards performance. However, turnover rates stayed fairly similar across performance groups, suggesting that compensation alone is not enough to retain top-performing employees.

Results Table & Chart:

Performance Rating	Min Salary Hike	Avg Salary Hike	Max Salary Hike	Total Employees	Attrition Count	Attrition Rate
3	11	14.00	19	1,244	200	16.08%
4	20	21.85	25	226	37	16.37%



Insights:

The analysis shows a clear increase in average salary hikes as performance ratings rise, confirming that the organization follows a strong pay-for-performance compensation model.

- Top-performing employees receive higher minimum and maximum salary increases, giving them greater earning and reward potential compared to lower-performing employees.

- Despite higher compensation for top performers, attrition rates remain nearly identical across performance levels, suggesting that salary increases alone are not enough to retain employees and that factors such as career growth, recognition, and job satisfaction also play a major role in retention.

Business Recommendations:

- Improve consistency in salary increase distribution to ensure performance-based rewards are applied more fairly across employees.
- Expand retention strategies beyond compensation by investing in career growth, recognition, and employee engagement programs.

QUERY 4: Income Distribution Among Attrition vs Retained Employees:

RESEARCH QUESTION: Does compensation decide who stays and who leaves?

SQL Query

```
SELECT
  Attrition,

  COUNT(*) AS total_employees,

  ROUND(AVG(MonthlyIncome), 2) AS avg_income,
  ROUND(AVG(JobLevel), 2) AS avg_job_level,
  ROUND(AVG(YearsAtCompany), 2) AS avg_years_at_company,
  ROUND(AVG(JobSatisfaction), 2) AS avg_job_satisfaction,

  ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM IBM), 2) AS employee_percent

FROM IBM
```

GROUP BY Attrition
ORDER BY Attrition;

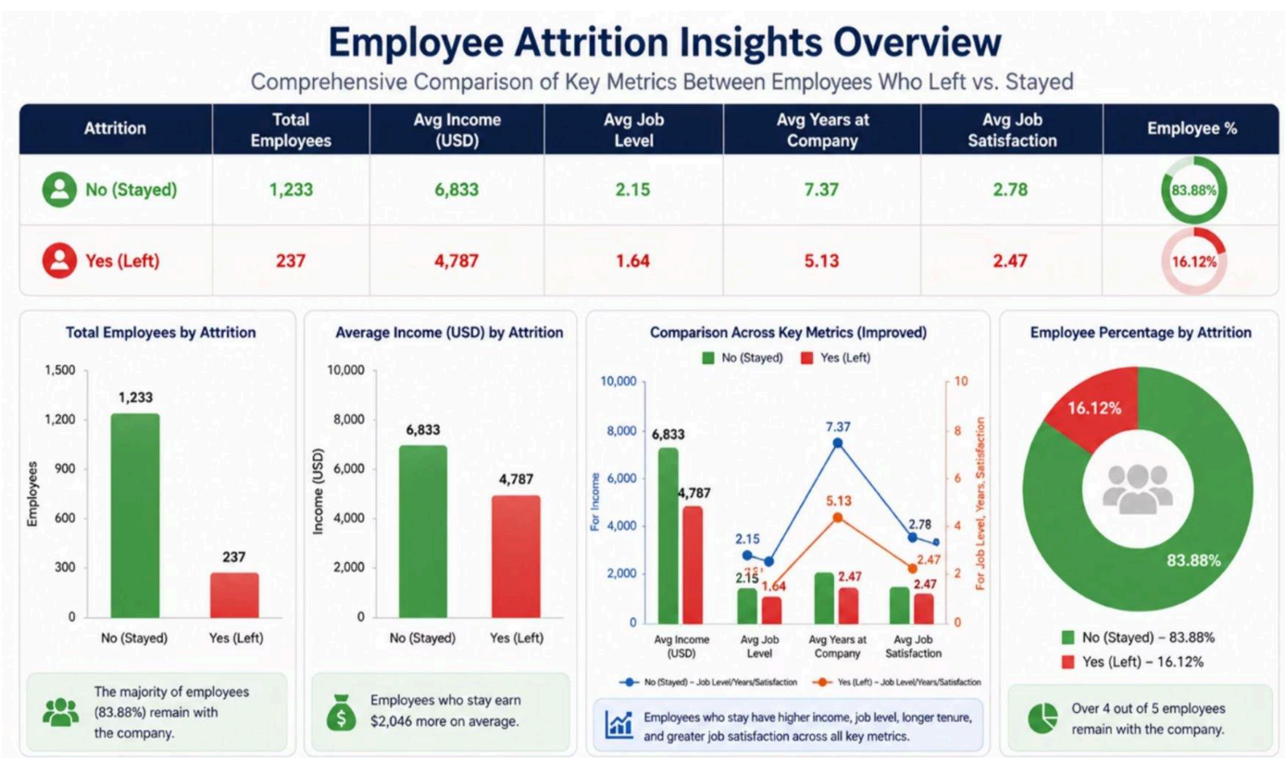
Explanation:

This query compares employees who remained with the company to those who left by analyzing income, job level, years at the company, and job satisfaction. The goal was to identify common patterns among employees with higher turnover risk. The results revealed that employees who left tended to earn lower salaries, hold lower-level positions, have shorter tenure, and report slightly lower satisfaction levels than employees who stayed.

Results Table:

Attrition	Total Employees	Avg Income	Avg Job Level	Avg Years at Company	Avg Job Satisfaction	Employee %
No	1,233	6,833	2.15	7.37	2.78	83.88%
Yes	237	4,787	1.64	5.13	2.47	16.12%

Charts:



Insights:

The analysis shows that employees who left the company earned significantly lower average incomes compared to employees who stayed, highlighting a strong connection between compensation and employee retention.

- Employees who left also tended to have lower job levels, shorter tenure, and slightly lower job satisfaction, suggesting that attrition is influenced by a combination of compensation, career growth opportunities, and overall employee experience.

- Although only 16.12% of employees left the company, the data shows a clear separation between retained and attrition groups, with departing employees being more concentrated in lower-income, lower-level, and early-career positions.

Business Recommendations:

- Increase compensation competitiveness for lower-income and early-career employees who face the highest attrition risk.
- Strengthen onboarding, mentorship, and career development programs to improve long-term employee retention.

QUERY 5: Stock Option Level vs Attrition

RESEARCH QUESTION: Does stock options play a role in retaining employees?

SQL Query

```
WITH total AS (
  SELECT StockOptionLevel, COUNT(*) AS total_employees
  FROM IBM
  GROUP BY StockOptionLevel
),
attrition AS (
  SELECT StockOptionLevel, COUNT(*) AS attrition_count
  FROM IBM
  WHERE Attrition = 'Yes'
  GROUP BY StockOptionLevel
)

SELECT
  t.StockOptionLevel,
  a.attrition_count * 100.0 / t.total_employees AS attrition_rate
FROM total t
LEFT JOIN attrition a
```

```
    ON t.StockOptionLevel = a.StockOptionLevel
ORDER BY t.StockOptionLevel;
```

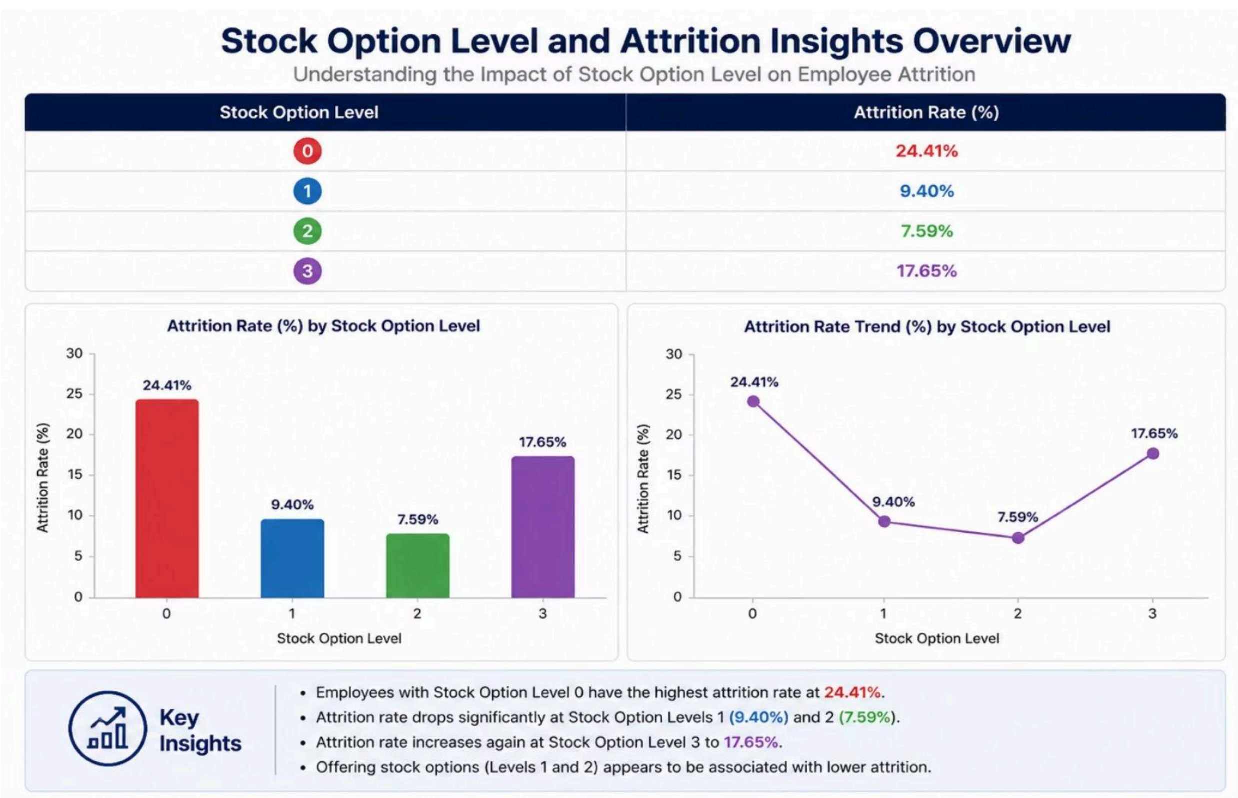
Explanation:

This query analyzes the relationship between stock option levels and employee turnover. It calculates attrition rates for each stock option group to determine whether equity compensation helps improve retention. The results showed that employees without stock options had the highest attrition rates, while employees with moderate stock option levels experienced much lower turnover, indicating that stock ownership can positively impact retention.

Results Table:

Stock Option Level	Attrition Rate (%)
0	**24.41**
1	**9.40**
2	**7.59**
3	**17.65**

Charts:



Insights:

The analysis shows that employees with no stock options experience the highest attrition rates, while employees with moderate stock option levels (Levels 1 and 2) have significantly lower attrition, suggesting that equity compensation is an effective retention incentive.

- Attrition rates decline consistently as stock option levels increase from 0 to 2, demonstrating a strong relationship between stock ownership and employee retention across most groups.
- The slight increase in attrition at Stock Option Level 3 suggests that senior-level employees may be influenced by factors beyond compensation alone, such as external opportunities, career advancement, and leadership mobility, highlighting the need for a broader retention strategy for high-level talent.

Business Recommendations:

- Expand stock option opportunities for employees in lower compensation groups to improve retention incentives.
- Develop broader retention strategies for senior employees that include leadership growth, career advancement, and executive engagement opportunities.

QUERY 6: Holistic Analysis – Full Attrition Profile

RESEARCH QUESTION: Who are the employees who left, what were the key driving factors?

SQL Query

```
SELECT
  CASE
    WHEN MonthlyIncome < (SELECT AVG(MonthlyIncome) FROM IBM)
    THEN 'Low Income'
    ELSE 'High Income'
 END AS income_group,

 CASE
   WHEN JobSatisfaction <= 2
   THEN 'Low Satisfaction'
   ELSE 'High Satisfaction'
```

```
END AS satisfaction_group,

CASE
  WHEN YearsAtCompany < 2
  THEN 'Early Tenure'
  ELSE 'Established Tenure'
    END AS tenure_group,

    COUNT(*) AS total_IBM,
    COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count,
    COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*) AS
  attrition_rate

  FROM IBM
  GROUP BY income_group, satisfaction_group, tenure_group
  ORDER BY attrition_rate DESC;
```

Explanation:

This query combines multiple employee factors, including income level, job satisfaction, and tenure, to identify the profiles most associated with turnover. Instead of looking at a single factor independently, the analysis evaluates how these conditions work together to influence attrition. The results showed that employees with low income, low satisfaction, and short tenure had the highest likelihood of leaving, while employees with higher income, stronger satisfaction, and longer tenure showed the lowest attrition risk.

Results Table:

Rank	Employee Profile	Attrition Rate (%)	Total Employees	Key Driver
1	Low Income • Low Satisfaction • Early Tenure	45.71	70	All 3 risk factors
2	Low Income • High Satisfaction • Early Tenure	35.64	101	Low income + early tenure
3	High Income • Low Satisfaction • Early Tenure	23.53	17	Low satisfaction + early tenure
4	Low Income • Low Satisfaction • Established Tenure	17.11	304	Low income + low satisfaction
5	High Income • Low Satisfaction • Established Tenure	13.48	178	Low satisfaction

6	Low Income • High Satisfaction • Established Tenure	12.95	502	Low income
7	High Income • High Satisfaction • Early Tenure	11.11	27	Early tenure
8	High Income • High Satisfaction • Established Tenure	7.75	271	No major risk factors

Charts:

Insights:

Employees with low income, low job satisfaction, and early tenure have the highest attrition rate at 45.71%, making this combination the strongest indicator of employee turnover risk.

- The results show that attrition rates decrease as individual risk factors are removed, demonstrating that higher income, stronger job satisfaction, and longer tenure each contribute to improved employee retention.

    - Employees with high income, high satisfaction, and established tenure have the lowest attrition rate at 7.75%, indicating that stable compensation, positive employee experience, and long-term organizational connection are key drivers of retention.

Business Recommendations:

- Prioritize retention initiatives for employees with low income, low satisfaction, and early tenure since they represent the highest-risk group.
- Implement proactive employee monitoring and engagement programs to identify and address attrition risks before turnover occurs.

SECTION - 4
CAREER & TENURE

Section Owner:

Divya

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Introduction

This section examines how career progression patterns, manager relationships, role fit, training investment, and tenure milestones relate to employee attrition. Five individual analyses build toward a composite risk segmentation model that identifies current employees most likely to leave.

Query 1: Attrition by Time Since Last Promotion

Stakeholder: HR Director — Career Progression & Retention Policy

Research Question: Does promotion recency predict attrition risk, and can promotion stagnation thresholds be used to identify at-risk employee segments before voluntary turnover occurs?

SQL Query

```
WITH CTE1 AS (
SELECT
CASE
        WHEN YearsSinceLastPromotion <=1 Then '0-1 yrs (Recent)'
        When  YearsSinceLastPromotion <= 3 Then '2-3 yrs (EarlyProgression)'
        When  YearsSinceLastPromotion <= 5 Then '4-5 yrs (MidProgression)'
        When  YearsSinceLastPromotion <= 10 Then '6-10 yrs (StagnationRisk)'
        Else '11+ yrs (HighStagnation)'
        End As PromotionRange,
CASE
       WHEN YearsSinceLastPromotion <= 1 THEN 1
       WHEN YearsSinceLastPromotion <= 3 THEN 2
       WHEN YearsSinceLastPromotion <= 5 THEN 3
       WHEN YearsSinceLastPromotion <= 10 THEN 4
       ELSE 5
    END AS SortOrder,
COUNT(*) AS TotalEmployees,
COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS EmployeeAttrition
FROM cleanedibm
GROUP BY PromotionRange, SortOrder
)
SELECT PromotionRange, TotalEmployees, EmployeeAttrition,
ROUND(EmployeeAttrition * 100.0 / TotalEmployees,2) AS AttritionRate
```

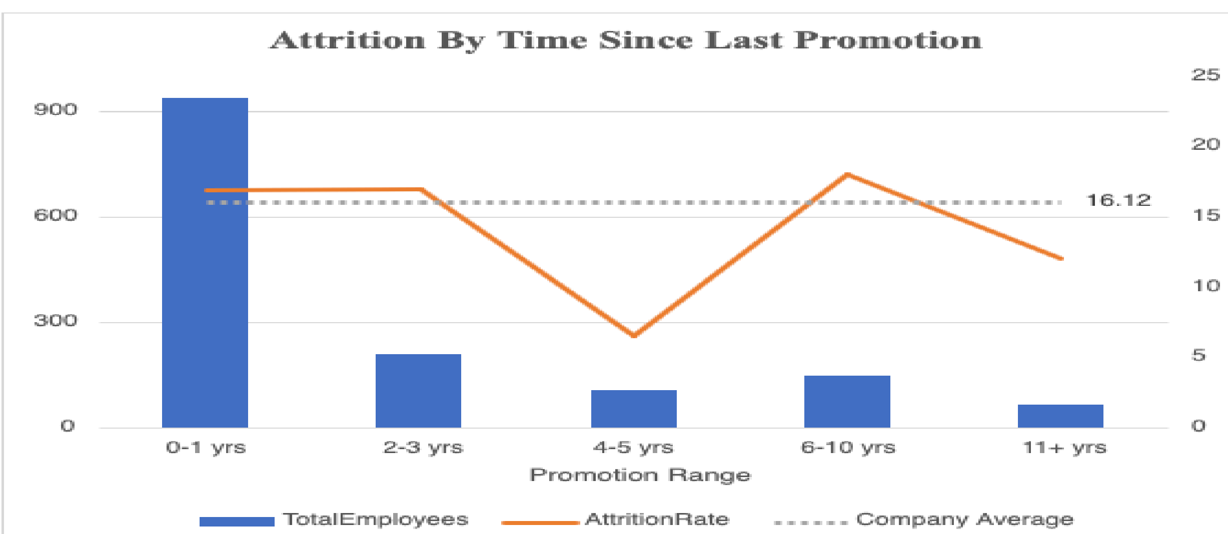
```
FROM CTE1
ORDER BY SortOrder;
```

Query Explanation This query uses a CTE to separate the two distinct responsibilities - bucketing logic from the final aggregation and rate calculation - for better readability. The CASE WHEN expression groups employees into five promotion recency bands based on YearsSinceLastPromotion.

A SortOrder column is included inside the CTE to enforce chronological ordering in the output — without it, MySQL would sort alphabetically, misrepresenting the progression of stagnation over time. It avoids the need for a secondary ORDER BY expression on the alias while ensuring chronological output regardless of alphabetical label names. The outer SELECT calculates AttritionRate by dividing EmployeeAttrition by TotalEmployees, keeping the rate calculation clean and separate from the grouping logic.

Result Table & Chart

Promotion Range	Total Employees	Employee Attrition	Attrition Rate (%)
0-1 yrs (Recent)	938	159	16.95
2-3 yrs (Early Progression)	211	36	17.06
4-5 yrs (Mid Progression)	106	7	6.60
6-10 yrs (Stagnation Risk)	149	27	18.12
11+ yrs (High Stagnation)	66	8	12.12



Key Insights

- Attrition peaks at **18.12%** in the 6-10 year band — the highest of all promotion groups and above the company average of 16.12%, confirming promotion stagnation as a measurable flight risk signal
- The **4-5 year window shows the lowest attrition at 6.60%** — employees in this stability phase feel settled and recognised, representing the optimal retention window
- Both the 0-1 year (16.95%) and 2-3 year (17.06%) bands show above-average attrition, suggesting that early-stage employees require role clarity and onboarding support independent of promotion timing
- The drop to 12.12% at 11+ years likely reflects adaptation rather than satisfaction — long-stagnated employees have fewer external options or have accepted their career trajectory

Business Recommendations: The HR Director should flag employees crossing the five year mark without promotion as a priority cohort for proactive career progression conversations. Introducing structured promotion review checkpoints at the five year milestone, combined with internal mobility pathways for mid-tenure employees, and improving **early-stage onboarding & role clarity** could meaningfully reduce attrition in the highest-risk stagnation band.

Query 2: Years with current manager vs attrition

Stakeholder: Department Manager

Research Question: Does the length of an employee's relationship with their current manager predict attrition, and does early instability in this relationship represent a measurable retention risk?

SQL Query

```
With CTE1 As (
Select

Case
        When YearsWithCurrManager <=1 Then '0-1 yrs (NewReln)'
        When  YearsWithCurrManager <= 3 Then '1-3 yrs (EarlyReln)'
        When  YearsWithCurrManager <= 5 Then '3-5 yrs (StableReln)'
        When  YearsWithCurrManager <= 10 Then '5-10 yrs (LongTermReln)'
        Else '10+ yrs (VeryLongTermReln)'
```

```
        END as ManagerTenure,

CASE
        When YearsWithCurrManager <=1 Then 1
         When YearsWithCurrManager <= 3 Then 2
         When YearsWithCurrManager <= 5 Then 3
         When YearsWithCurrManager <= 10 Then 4
          ELSE 5
        END AS SortOrder,

   Attrition
   From cleanedibm
   )

Select ManagerTenure,
COUNT(*) AS TotalEmployees,
COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS AttritionCount,
ROUND(
COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*),
2) AS AttritionRate

From CTE1
Group BY ManagerTenure, SortOrder
Order By SortOrder;
```

Query Explanation:

In this query CTE is used to pull individual employee rows with their bucketed ManagerTenure label and SortOrder, along with the raw Attrition value. It retains row-level data so the outer SELECT has full flexibility to aggregate as needed.

The outer SELECT then performs all aggregation — counting total employees, counting attrition cases, and calculating the rate.

This two-layer structure separates the bucketing logic from the counting logic, making each step independently readable. The conditional COUNT approach — COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) — avoids the need for a separate filtered subquery and produces attrition counts and rates in a single pass through the data.

Result Table & Chart

Employee Attrition Rate By Manager Tenure

ManagerTenure	Total Employees	AttritionRate	AvgAttrition	Attrition Count
0 - 1 yrs	339	28.32	16.12	96
2 -3 yrs	486	14.2	16.12	69
4 - 5 yrs	129	11.63	16.12	15
6 - 10 yrs	443	12.19	16.12	54
11+ yrs	73	4.11	16.12	3

Key Insights:

- Attrition is **highest at 28.32% in the first year of a manager relationship** — nearly double the company average of 16.12% — confirming that early manager-employee misalignment is a primary retention risk
- **Attrition declines consistently and significantly with manager tenure** — from 28.32% at 0-1 years to just 4.17% at 11+ years, a 24 percentage point reduction representing the single largest attrition gradient observed across all analyses
- Employees who reach the 4-5 year mark with the same manager show 13.08% attrition — below the company average — suggesting that once trust and working rhythm are established, the relationship becomes a retention asset rather than a risk factor

Recommendations:

- Strengthen **manager–employee alignment during onboarding**
- Introduce **early feedback checkpoints (first 6–12 months)**
- Train managers on **first-year employee engagement**
- Use **early-stage attrition as a signal of manager effectiveness** to guide leadership development

Query 3: Training Investment vs Employee Attrition

Stakeholder: L&D Manager — Learning Investment & Development Strategy

Research Question: Does the frequency of training sessions received predict employee attrition, and does zero training represent a measurable and actionable retention risk signal?

SQL Query

```
SELECT
   TrainingTimesLastYear,
   COUNT(*) AS TotalEmployees,
   COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS EmployeeAttrition,
   ROUND(
      COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*),
   2) AS AttritionRate
FROM cleanedibm
GROUP BY TrainingTimesLastYear
ORDER BY TrainingTimesLastYear;
```

Query Explanation: This query groups employees by the number of training sessions attended in the last year — TrainingTimesLastYear — and calculates the attrition count and rate for each value. TrainingTimesLastYear is a discrete integer from 0 to 6, making each value a meaningful and distinct category. The conditional COUNT approach calculates attrition within each group without requiring a subquery or JOIN.

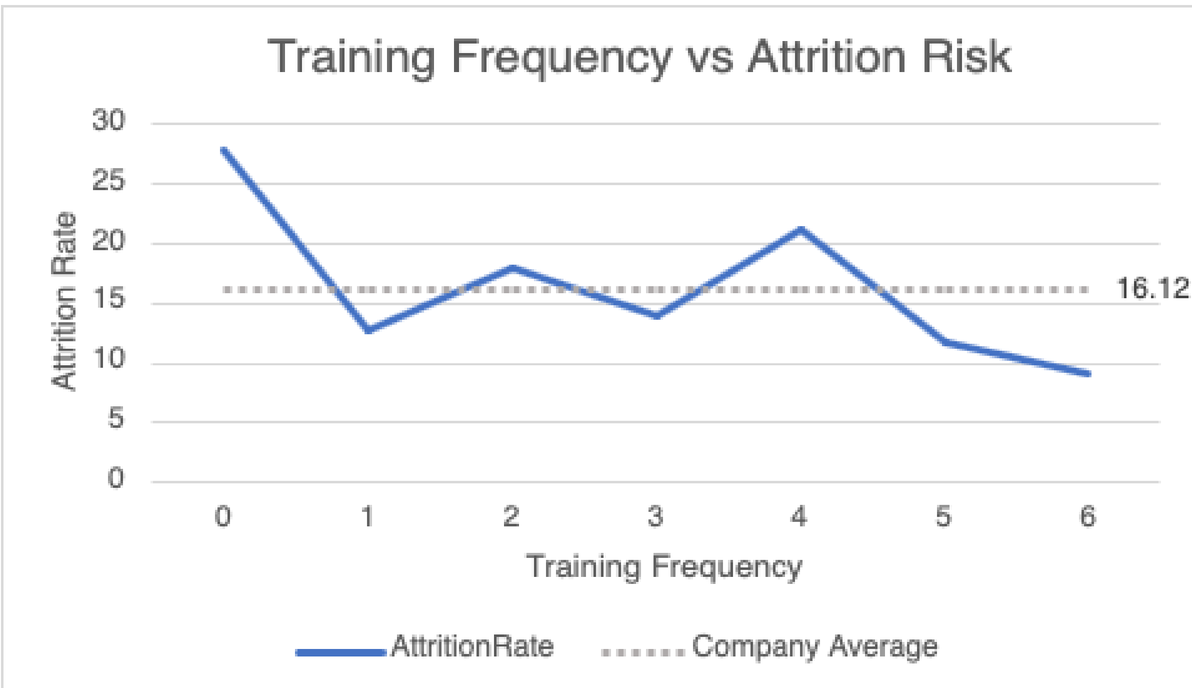
Results are ordered by training frequency to reveal the directional relationship between investment on training and attrition.

Result Table

TrainingTimesLastYear	Total Employees	Employee Attrition	AttritionRate (%)
0	54	15	27.78
1	71	9	12.68
2	547	98	17.92

3	491	69	14.05
4	123	26	21.14
5	119	14	11.76
6	65	6	9.23

Chart:



Key Insights

- **Employees who received zero training sessions show 27.78% attrition — the highest of all groups and nearly double the company average of 16.12% —** confirming that absence of training investment is the strongest training-related attrition signal in this dataset
- Attrition **generally declines as training frequency increases** — employees receiving six sessions show the lowest attrition at 9.23%, less than a third of the zero-training rate
- The relationship is not perfectly linear — there is an unexpected spike at four sessions (21.14%) before the trend resumes downward. This blip may reflect intensive development of high-potential employees who subsequently receive

external offers, and should be treated as a directional caveat rather than a contradictory finding
- The pattern at the extremes is clear and actionable: no training doubles risk, regular training nearly halves it

Business Recommendation: The L&D Manager should mandate a minimum of one training session annually for all employees — zero training is the single most actionable training-related attrition lever identified in this analysis. Employees in the Stagnation Risk promotion band identified in Query 1 should be prioritised for targeted development programmes, as compounding risk factors increase the urgency of intervention.

Query 4: Attrition Risk by Tenure Milestone

Stakeholder: HR Director — Onboarding & Lifecycle Retention Strategy

Research Question: At which stage of an employee's company journey is attrition risk highest, and can tenure milestones be used to time retention interventions more effectively?

SQL Query

```
WITH TenureBuckets AS (
  SELECT
    CASE
      WHEN YearsAtCompany <= 1 THEN '0-1 yrs (New Hires)'
      WHEN YearsAtCompany <= 3 THEN '2-3 yrs (Early Stage)'
      WHEN YearsAtCompany <= 5 THEN '4-5 yrs (Settled Stage)'
      WHEN YearsAtCompany <= 10 THEN '6-10 yrs (Long Term)'
      ELSE '11+ yrs (Very Long Term)'
    END AS TenurePhase,
    CASE
      WHEN YearsAtCompany <= 1 THEN 1
      WHEN YearsAtCompany <= 3 THEN 2
      WHEN YearsAtCompany <= 5 THEN 3
      WHEN YearsAtCompany <= 10 THEN 4
      ELSE 5
    END AS SortOrder,
    Attrition
  FROM cleanedibm
)
```

```
SELECT
  TenurePhase,
  COUNT(*) AS TotalEmployees,
  COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS EmployeeAttrition,
  ROUND(
    COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) * 100.0 / COUNT(*),
  2) AS AttritionRate
FROM TenureBuckets
GROUP BY TenurePhase, SortOrder
ORDER BY SortOrder;
```

Query Explanation: This query uses a single CTE to bucket employees by YearsAtCompany into five career stage phases and assign a SortOrder for chronological output.

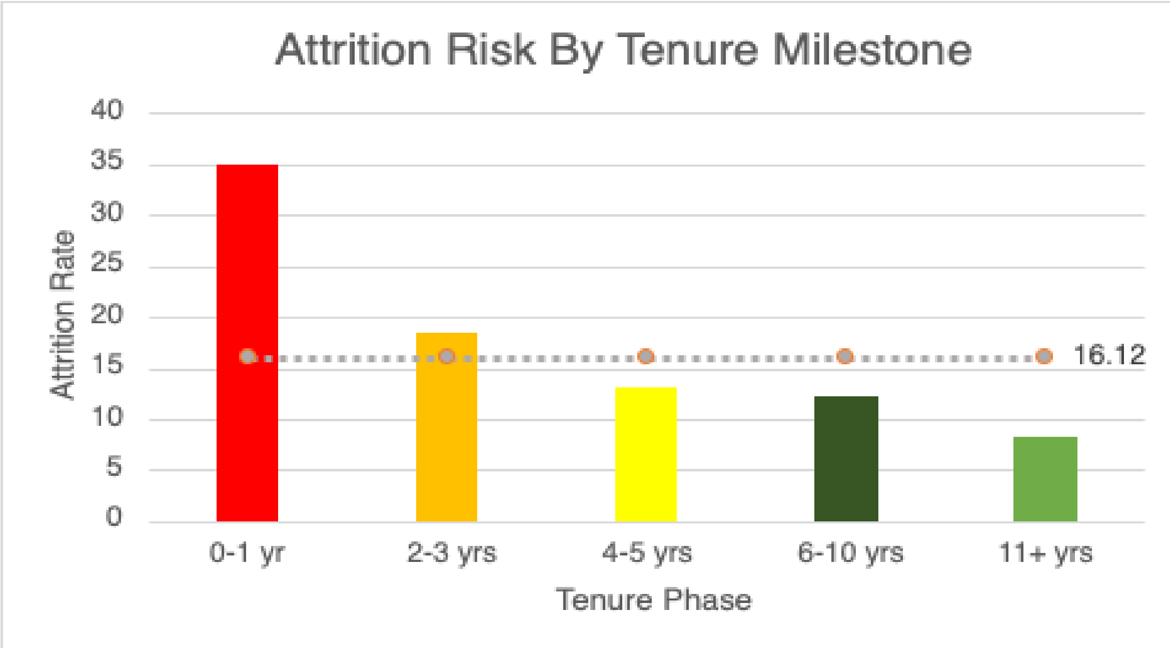
The CTE retains the raw Attrition column so the outer SELECT can perform conditional counting. Labels were chosen to reflect the organisational meaning of each tenure phase — New Hires, Early Stage, Settled Stage, Long Term and Very Long Term — rather than using generic numeric ranges, making results directly readable by HR stakeholders without requiring additional interpretation.

The CTE approach was chosen over a direct GROUP BY CASE WHEN to allow SortOrder to be defined once and reused in the ORDER BY clause without repeating the full CASE WHEN expression.

Result Table

Tenure Phase	Total Employees	Employee Attrition	Attrition Rate (%)
0-1 yrs (New Hires)	215	75	34.88
2-3 yrs (Early Stage)	255	47	18.43
4-5 yrs (Settled Stage)	306	40	13.07
6-10 yrs (Long Term)	448	55	12.28
11+ yrs (Very Long Term)	246	20	8.13

Chart:



Key Insights

- **1 in 3 new hires leaves within their first year** — a 34.88% attrition rate more than double the company average of 16.12% — making the first twelve months the highest single risk window identified across all five Career & Tenure analyses
- **Attrition declines consistently and significantly with tenure** — from 34.88% in year one to 8.13% at 11+ years, a clean downward trend with no reversals, confirming tenure as the strongest retention stabiliser in this dataset
- Employees who reach the 4-5 year mark show below-average attrition at 13.07% — suggesting that crossing the three year threshold represents a meaningful loyalty inflection point
- Very long term employees (11+ years) leave at just 8.13% — less than a quarter of the new hire rate — confirming that investment in early retention compounds significantly over the employee lifecycle

Business Recommendation: The HR Director should prioritise structured onboarding programmes with 30-60-90 day check-ins and early manager alignment conversations specifically designed to move new hires through the critical first year threshold. Every percentage point reduction in first-year attrition has compounding value — employees who reach four years become significantly more stable and require substantially less retention investment.

Query 5: Identifying At-Risk Employees Before They Leave — Holistic Risk Segmentation

Stakeholder: CEO / Board of Directors — Strategic Workforce Risk Management

Research Question: Can multiple attrition risk factors identified across all domain analyses be combined into a single composite score to identify current employees most likely to leave, enabling proactive intervention before resignation occurs?

SQL Query

```
WITH RiskScoring AS (
  SELECT
    EmployeeNumber,
    JobRole,
    Department,
    YearsAtCompany,
    MonthlyIncome,
    (
      CASE WHEN YearsAtCompany <= 1 THEN 2 ELSE 0 END +
      CASE WHEN JobSatisfaction <= 2 THEN 2 ELSE 0 END +
      CASE WHEN YearsSinceLastPromotion > 5 THEN 2 ELSE 0 END +
      CASE WHEN YearsWithCurrManager < 2 THEN 1 ELSE 0 END +
      CASE WHEN TrainingTimesLastYear < 2 THEN 1 ELSE 0 END +
      CASE WHEN WorkLifeBalance <= 2 THEN 1 ELSE 0 END +
      CASE WHEN EnvironmentSatisfaction <= 2 THEN 1 ELSE 0 END
    ) AS RiskScore
  FROM cleanedibm
  WHERE Attrition = 'No'
),
Segmented AS (
  SELECT *,
    CASE
      WHEN RiskScore >= 6 THEN 'Critical Risk'
      WHEN RiskScore >= 4 THEN 'High Risk'
      WHEN RiskScore >= 2 THEN 'Moderate Risk'
      ELSE 'Low Risk'
    END AS RiskSegment,
    CASE
      WHEN RiskScore >= 6 THEN 1
```

```
            WHEN RiskScore >= 4 THEN 2
            WHEN RiskScore >= 2 THEN 3
            ELSE 4
        END AS SortOrder
    FROM RiskScoring
)
SELECT
    RiskSegment,
    COUNT(*) AS EmployeeCount,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Segmented), 1) AS PercentAtRisk,
    ROUND(AVG(MonthlyIncome), 0) AS AvgMonthlyIncome,
    ROUND(AVG(YearsAtCompany), 1) AS AvgTenure
FROM Segmented
GROUP BY RiskSegment, SortOrder
ORDER BY SortOrder;
```

Query Explanation: This is the most complex query in this analysis, using a chained two-CTE structure to perform scoring, segmentation, and aggregation in three distinct layers.

The first CTE — RiskScoring — filters to current employees only (Attrition = 'No') and calculates a composite RiskScore for each individual by summing binary CASE WHEN conditions across seven risk factors. Each condition contributes a weighted point value based on the strength of evidence from this analysis.

The second CTE — Segmented — takes each scored employee and applies a second CASE WHEN to translate the numeric score into a human-readable risk tier.

The final SELECT aggregates employees by tier and calculates average income and tenure as profile descriptors for each segment. The total percentage calculation uses a subquery — SELECT COUNT(*) FROM Segmented — as a denominator, substituting for a Window Function which MySQL does not support inside GROUP BY aggregations.

This architecture also makes the query maintainable — if risk weights need adjustment, only the RiskScoring CTE changes. If tier thresholds need updating, only Segmented changes. The scoring weights of 2 assigned to three factors — new hire tenure, job satisfaction, and promotion stagnation — are evidence-based: these three conditions showed the highest individual attrition rates in Queries 1, 4, and 5 respectively.

The risk factor was calculated as a weighted score using the factors and scores below:

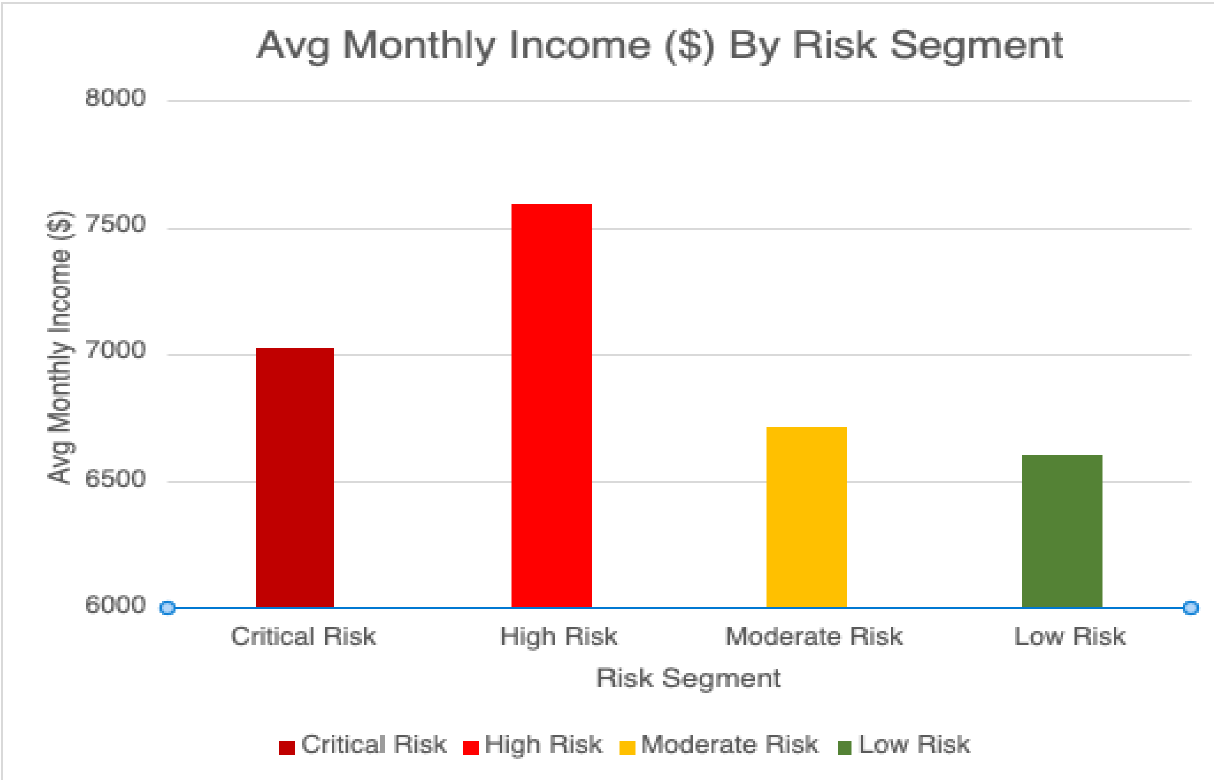
Risk Factor Weighting Rationale

Risk Factor	Column	Condition	Weight	Evidence
New hire tenure	YearsAtCompany	<= 1 year	+2	34.88% attrition — Query 5
Low job satisfaction	JobSatisfaction	<= 2	+2	Strongest satisfaction predictor
Promotion stagnation	YearsSinceLastPromotion	> 5 years	+2	18.12% attrition — Query 1
New manager relationship	YearsWithCurrManager	< 2 years	+1	28.32% attrition — Query 2
Low training investment	TrainingTimesLastYear	< 2 sessions	+1	27.78% zero-training rate — Query 4
Poor work-life balance	WorkLifeBalance	<= 2	+1	Bottom half of 1-4 survey scale
Low environment satisfaction	EnvironmentSatisfaction	<= 2	+1	Workplace dissatisfaction signal

Result Table

Risk Segment	Employee Count	% of Workforce	Avg Monthly Income	Avg Tenure (yrs)
Critical Risk	42	3.4%	$7,021	5.3
High Risk	211	17.1%	$7,594	7.3
Moderate Risk	518	42.0%	$6,713	7.8
Low Risk	462	37.5%	$6,602	7.1

Chart:



Avg Monthly Income ($) By Risk Segment
8000
7500
7000
6500
6000
Avg Monthly Income ($)
Critical Risk
High Risk
Moderate Risk
Low Risk
Risk Segment
Critical Risk
High Risk
Moderate Risk
Low Risk

Employees By Risk Segment
42
211
462
518
Critical Risk
High Risk
Moderate Risk
Low Risk

Key Insights

- **253 current employees — 1 in 5 of the active workforce — fall into High or Critical Risk segments**, representing an immediately actionable retention priority for the CEO and HR leadership team
- **High Risk employees earn above the company average at $7,594 monthly income** — these are not disengaged low performers but relatively well-compensated employees whose dissatisfaction is rooted in stagnation, manager relationships, and development gaps rather than compensation. Salary adjustments alone will not retain them
- The **Moderate Risk segment of 518 employees (42% of workforce) represents a silent disengagement pipeline** — if compounding risk factors are left unaddressed, this group is the most likely source of future Critical and High Risk escalations
- Critical Risk employees average just 5.3 years tenure — experienced enough to have significant institutional knowledge but early enough in their career to have strong external options. This combination makes them both high-value and high-flight-risk simultaneously

Business Recommendation: The CEO and Board should implement a quarterly People Risk Review in which HR Directors use this composite scoring framework to flag Critical and High Risk employees for targeted retention conversations covering promotion timelines, manager relationship quality, and development investment. Acting proactively on 253 identified employees is significantly less costly than replacing them — the average cost of employee replacement is estimated at 50-200% of annual salary. The 518 Moderate Risk employees should be monitored as an early warning pipeline, with light-touch engagement interventions such as development conversations and satisfaction check-ins prioritised before risk scores escalate.

Strategic Recommendations — What the Data Tells Us

A Cross-Domain Retention Framework for IBM HR Leadership

Across 20+ analyses spanning compensation, satisfaction, career progression, and tenure, one pattern is consistent — attrition is not sudden. The signals accumulate across months and years: a missed promotion, a new manager, no training, a role that stopped growing. The composite risk model developed in this analysis demonstrates that these signals are measurable, predictable, and actionable.

With the right data infrastructure and a structured quarterly review process, this organisation has the tools to move from reactive replacement to proactive retention.

HIGHEST RISK WINDOW **34.88%** New hire attrition — 1 in 3 leave within year one	HIGHEST RISK PROFILE **45.71%** Low income · low satisfaction · early tenure combined
EMPLOYEES FLAGGED **253** Current staff in High or Critical Risk segments	OVERALL ATTRITION RATE **16.1%** Company-wide baseline this analysis

Conclusion: This analysis set out to understand why 16.1% of IBM's workforce chose to leave — and what the data reveals is that attrition is rarely sudden. Across compensation, satisfaction, career progression, and tenure, the same pattern emerges: risk accumulates quietly through missed promotions, new manager transitions, absent training, and roles that stop growing. The value of a data-driven approach lies not in explaining departures after the fact, but in identifying the signals early enough to act. The composite risk model developed in this report demonstrates that with the right analytical framework, organisations can shift from reactive replacement to proactive retention — protecting both their people and their bottom line. Through our analysis, we have identified the key risk factors and thus, have consolidated a cross-domain recommendations table for the stakeholders and decision-makers.

Cross-Domain Recommendations Table

Priority	Finding	Stakeholder	Recommendation
1	1 in 3 new hires leaves in year one	HR Director	Structured 30-60-90 day onboarding with manager alignment
2	Overtime doubles attrition at every satisfaction level	COO	Treat overtime reduction as highest ROI retention lever
3	Employees with zero stock options leave at 24.41%	CFO	Expand equity access to lower compensation bands
4	Promotion stagnation beyond 5 years peaks at 18.12%	HR Director	Introduce promotion checkpoints at 5 year milestone
5	Low satisfaction + early tenure = 45.71% attrition	CEO/Board	Prioritise pulse surveys as early warning system
6	253 current employees flagged as High/Critical Risk	CEO/Board	Implement quarterly People Risk Review
7	Single employees leave at 26.9% — highest demographic	DEI Lead	Design retention programmes for single employees
8	Zero training employees leave at 27.78%	L&D Manager	Mandate minimum one training session annually

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